One Fed, Many Voices: Coordinated Communication vs. Transparent Debate*

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Abstract

We study how coordination in central bank communication affects monetary policy transmission. Using FOMC speeches, we measure similarity to the Chair's preceding post-meeting communication as an observable manifestation of coordination and construct speech surprises from high-frequency changes in eurodollar futures. On average, these surprises have little effect on inflation expectations or equity prices. Greater coordination strengthens transmission: aligned speeches are not redundant for financial markets but intensify earlier policy signals, lowering inflation expectations and equity prices after tightening surprises. At lower similarity, responses weaken and may reverse. A nonlinear general equilibrium model with incomplete information rationalizes these findings.

JEL codes: E52; E58; G14; C55; D83.

Keywords: Monetary policy transmission, text analysis, Cosine similarity, Large language models, FOMC speeches, High-frequency identification.

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1 Introduction

In the United States—as in many other countries—monetary policy is set by a committee, the Federal Open Market Committee (FOMC), which brings together the Board of Governors and the presidents of the 12 Federal Reserve Banks. Both governors and Reserve Bank presidents regularly deliver public speeches conveying their views on the economic outlook and the appropriate stance of policy. At times, these remarks closely reflect the Committee's consensus; at others, they reveal a broader range of individual perspectives. Since speakers' intentions are not observed by markets, we interpret similarity as an observable manifestation of coordination. We therefore leverage variation in speech alignment to study the quantitative role of coordination in FOMC communication for monetary policy transmission.

We assemble a dataset of speeches delivered by FOMC participants since 2011. Each speech is time-stamped to the minute, allowing us to match it with five-minute eurodollar futures prices and construct the associated high-frequency "monetary policy speech" (MPS) surprise. We measure the verbal alignment of each speech with the Chair's most recent post-meeting press conference using *cosine similarity*, a standard text-based metric in natural language processing. Our baseline measure uses the full press conference, including both the Chair's prepared remarks and the subsequent Q&A.

There is a key asymmetry in interpreting similarity as coordinated communication. Greater similarity can be interpreted as an observable manifestation of coordination: because financial markets cannot distinguish deliberate from incidental alignment, our analysis need not distinguish between the two. Lower similarity, however, need not imply disagreement or a lack of coordination, as speeches may simply discuss different topics or convey additional information.

Our empirical analysis proceeds in two steps. First, we estimate daily local projections of Treasury yields, breakeven inflation, and equity prices on MPS surprises to characterize their average effects. Second, we interact these surprises with our measure of speech similarity to the Chair's preceding press conference, allowing monetary policy transmission to vary with the degree of communication alignment.

Three key findings emerge from our analysis. First, on average, before accounting for differences in textual alignment with the preceding Chair's press conference, monetary policy surprises associated with FOMC participants' post-meeting speeches have no statistically significant effect on inflation expectations or equity prices. Second, when communication is more coordinated, monetary policy transmission becomes stronger: a tightening surprise raises short-term yields while lowering inflation expectations and equity prices. Thus, coordinated communication is not redundant: markets learn from subsequent aligned speeches,

which intensify the effects of the earlier policy signal. Third, when speeches are less aligned with the Chair's message, this pattern weakens substantially. Inflation expectations and equity prices become less responsive and can even move in the opposite direction.

We conduct an extensive set of exercises to assess the robustness and interpretation of these findings. First, to address potential endogeneity in our similarity measure, we remove the variation in similarity associated with intervening macroeconomic news arriving between the preceding FOMC press conference and subsequent speeches. The results are virtually unchanged, indicating that our findings are not driven by policymakers adjusting their communication in response to incoming economic data. Second, because cosine similarity relies on word overlap, we construct an LLM-based measure of semantic similarity that captures similarity in meaning rather than wording. This measure yields the same qualitative results, indicating that our findings are not driven by differences in rhetorical style or phrasing. Third, the results are also robust to alternative event windows, excluding Chair speeches, restricting the sample to FOMC participants with voting rights, considering only speech–press-conference pairs that discuss the same topic and agree, constructing similarity using FOMC-relevant content from contemporaneous deliberations, and excluding the Q&A portion of the Chair's press conference. Fourth, our findings remain robust to controlling for the effective lower bound, the time elapsed since the preceding press conference, speakers' institutional roles, and nonlinearities in monetary policy surprises, as well as to winsorizing extreme monetary policy surprises.

Finally, to interpret these findings, we develop an imperfect-information New Keynesian model in which the Chair's communication and subsequent FOMC speeches provide public signals about future monetary policy. Importantly, we solve the model globally in its nonlinear specification, allowing the effects of communication to depend on the similarity of these signals. The model isolates two mechanisms. First, in the baseline, similarity governs how precisely agents infer a common policy message. When the Chair and an FOMC participant convey similar messages, agents infer future policy more precisely, strengthening the transmission of an anticipated monetary tightening and generating a larger decline in inflation. Second, to account for the empirical finding that low-similarity speech surprises can raise both expected interest rates and inflation, we allow speeches to convey information not only about future policy but also about future demand. At low similarity, markets may interpret the additional content of a speech partly as information about future economic fundamentals rather than as reinforcement of the common policy signal. This mechanism can overturn the inflation response: A communication surprise that raises expected interest rates can therefore raise expected inflation when similarity is low, while lowering it when similarity is high. The model therefore rationalizes both sides of our empirical findings: why

coordinated communication strengthens monetary policy transmission and why less aligned speeches can generate qualitatively different market responses.

Literature Review. This paper contributes to the growing literature on central bank communication. Early contributions emphasize that communication is an important monetary policy instrument because it shapes private expectations and affects the transmission of policy decisions (Woodford, 2005; Blinder et al., 2008).

Our empirical approach builds on a large literature that uses high-frequency financial-market reactions around central-bank communication events to identify monetary-policy news and study its transmission. This approach was pioneered by Kuttner (2001) and Cochrane and Piazzesi (2002). Gürkaynak et al. (2005) use intraday asset-price movements around FOMC policy announcements to identify separate surprises about the current federal funds rate target and the expected future path of policy. Campbell et al. (2012) distinguish between Odyssean forward guidance about the future path of policy and Delphic communication that also reveals information about economic fundamentals. A subsequent literature develops high-frequency identification strategies that use the joint response of different asset prices, together with other information, to distinguish monetary-policy news from information about economic fundamentals (Campbell et al., 2017; Melosi, 2017; Nakamura and Steinsson, 2018; Cieslak and Schrimpf, 2019; Jarociński and Karadi, 2020; Andrade and Ferroni, 2021; Bauer and Swanson, 2023a; Miranda-Agrippino and Ricco, 2021). More recently, Gorodnichenko et al. (2023) show that the vocal tone of the Fed Chair during post-meeting press conferences contains information that affects financial markets.

More closely related to our focus on speeches, several papers study the information conveyed by policymakers outside scheduled policy announcements. Swanson (2023) and Swanson and Jayawickrema (2024) show that speeches and testimonies are important sources of monetary-policy news. Granziera et al. (2025) study how FOMC speeches shape households' and professional forecasters' inflation expectations, distinguishing information about inflationary pressures (Delphic) from communication about the Fed's intended policy response (Odyssean). Altavilla et al. (2026) use high-frequency financial-market movements around a broad set of euro-area policy events, including policymakers' speeches, to identify monetary-policy shocks and study their transmission. Acosta et al. (2026) construct a comprehensive high-frequency event-study database covering multiple forms of FOMC communication, including speeches. Our angle is to study the degree of coordination across speeches and show that it matters for financial markets.

Our measure of communication alignment relates to the growing use of textual methods in central banking. Hansen and McMahon (2016b) use computational linguistics to quan-

tify different dimensions of FOMC communication, while Hansen et al. (2018) use textual analysis to study transparency and deliberation within the FOMC. Ahrens and McMahon (2021) use supervised topic modeling to infer signals about GDP, inflation, and unemployment from central-bank speeches and construct a measure of dispersion across policymakers. Handlan (2022b) uses machine learning to construct text-based monetary-policy shocks that isolate the effects of forward guidance in FOMC statements from information about current economic conditions. Aruoba and Drechsel (2026) use pre-meeting Federal Reserve staff documents to capture policymakers' information set when identifying monetary-policy shocks. Handlan (2022a) use cosine similarity to quantify changes in wording across consecutive FOMC post-meeting statements and relate them to revisions in expected future federal funds rates around FOMC announcements. We instead compare individual policymakers' speeches with the Chair's preceding post-meeting press conference to measure coordination across FOMC voices, and study how it shapes the response of interest rates, inflation expectations, and equity prices to the speeches themselves.

A related literature studies heterogeneity within monetary-policy committees and the communication and behavior of individual policymakers. Ehrmann and Fratzscher (2007) compare communication strategies across central banks and characterize the Federal Reserve as combining collegial decision-making with relatively individualistic communication by committee members. Ehrmann and Fratzscher (2013) study dispersion in communication across committee members and its implications for the predictability of monetary policy, while Bordo and Istrefi (2023) document substantial heterogeneity in the perceived policy preferences of FOMC participants. Hansen and McMahon (2016a) study how signaling incentives shape individual monetary-policy committee members' behavior. Vissing-Jorgensen (2019) studies how policymakers' incentives to influence market expectations can generate competition for attention and a multiplicity of voices in Federal Reserve communication. Turning specifically to public speeches, Neuhierl and Weber (2019) show that the tone of FOMC speeches contains information about the future path of monetary policy, while Ehrmann et al. (2021) document that voting rights affect FOMC participants' communication behavior and the market response to their speeches. Malmendier et al. (2021) relate policymakers' inflation experiences to their voting behavior and the hawkishness of their speeches, while Istrefi et al. (2023) use speeches to measure policymakers' financial-stability concerns and relate them to monetary-policy decisions. The heterogeneity documented in this literature is the starting point of our analysis. We study how markets respond when this heterogeneity is reduced by greater coordination in policymakers' communication.

2 Data Sources

Our analysis combines three textual sources on Federal Reserve communications with macroeconomic release and forecast data. We structure the textual sources to capture the timing, authorship, and content of Fed communications, and use the macroeconomic data to measure news arriving between each press conference and subsequent speeches.

FOMC Press Conferences. Beginning in 2011, the Fed Chair has held press conferences following FOMC meetings. We obtain the full text of these conferences from the same Federal Reserve archive from April 2011 until November 2023. We parse each document to isolate individual questions and responses, generating a dataset that includes the date, the text of each journalist's question, and the corresponding reply from the Chair. This allows us to quantify the tone, content, and structure of real-time public communication by the Fed.

Public Speeches by FOMC Participants. We also compile a corpus of public speeches by individual FOMC participants—including Board members and Presidents of the twelve regional Federal Reserve Banks—delivered since 2011. The majority of these speeches are drawn from the FRASER archive (link), with additional documents obtained directly from Reserve Bank websites (e.g., Minneapolis Fed). Where available, we record the exact start and end times of speeches. In cases where end times are not reported, we impute them based on speech length and an estimated words-per-minute rate to ensure comparability across events. The public speech dataset is matched to the press-conference sample, so we consider speeches delivered between April 2011 and November 2023.

FOMC Meeting Transcripts. Finally, we collect a corpus covering internal FOMC deliberations. Since 2007, the Federal Reserve has published verbatim transcripts of FOMC meetings with a five-year lag. We collect these transcripts from the Federal Reserve Board's official archive and convert them from PDF into a structured format.[1] For each meeting, we extract the date, identify individual speakers using a harmonized naming convention, and record the full text of their interventions. This dataset enables speaker-level analysis of deliberation dynamics within the FOMC.

Macroeconomic News. We obtain macroeconomic releases and pre-release median forecasts from the Haver Analytics/MMS database. The indicators comprise the advance estimate of real GDP growth, headline and core CPI inflation, PCE inflation, nonfarm payrolls,

[1] The original transcripts are available at this link.

the unemployment rate, average hourly earnings, and industrial production. For each release, we measure the surprise as the first reported value minus the median forecast, expressed in the indicator's original units. We associate each speech with all releases occurring after the preceding FOMC press conference and before the start of the speech.

2.1 Text Similarity between Individual and Centralized Communication

To quantify the alignment between individual and centralized communications of the Federal Reserve, we construct a measure of similarity between the speeches given by individual FOMC members and those delivered by the Fed Chair in the most recent FOMC press conference.

We begin by standardizing the raw textual data: (i) converting all text to lowercase and removing punctuation and non-alphanumeric characters; (ii) lemmatizing words to their base forms to account for inflectional variation; (iii) removing common stopwords using the standard English dictionary from the Natural Language Toolkit (NLTK); and (iv) manually harmonizing the names of FOMC participants across datasets to ensure consistent speaker identification.

We then represent each text in the corpus numerically using the Term Frequency–Inverse Document Frequency (TF-IDF) method. This assigns a weight to each word in a document based on two components: term frequency, which reflects how often a word appears in a given document, and inverse document frequency, which downweights words that are common across many documents and thus less informative. The resulting TF-IDF vectors provide a high-dimensional representation of each statement that captures its semantic content.

Once statements are vectorized, we compute pairwise cosine similarity scores to quantify the textual alignment between individual FOMC participants speeches and the Fed Chair preceding press conference. Cosine similarity captures the angular distance between two TF-IDF vectors, with values closer to one indicating greater semantic similarity. We compare each FOMC member speech to the complete set of the Chair's remarks—including both prepared comments and responses to journalists—thereby capturing alignment with the totality of the Chair's public communication. This procedure yields a FOMC member-level similarity score for each meeting.

The total number of pairwise textual alignments (cosine similarity scores) between FOMC member speeches and the preceding press conference is 922.

2.2 Illustrative Examples and Dispersion across FOMC Members

To provide intuition for the cosine similarity scores computed in our analysis, we present a set of examples with particularly high values. Table 1 reports excerpts from selected FOMC speeches and the corresponding Chair's press conference remarks with which they exhibit the greatest similarity. These examples help validate that high similarity scores reflect substantive alignment in both language and content.

In the top-ranked case, Stanley Fischer's February 2016 speech closely mirrors Janet Yellen's December 2015 press conference, with a cosine similarity score of 0.81. Both emphasize the decision to raise the federal funds rate, the continued accommodative stance of monetary policy, and confidence in inflation returning to target. The alignment is not only thematic but also lexical, with near-identical phrasing across several statements.

Other high-similarity examples demonstrate similar patterns. Powell's 2013 speech reiterates Bernanke's emphasis on subdued inflation and the persistence of accommodative financial conditions. Likewise, speeches by Fischer (2015) and Brainard (2017) echo the Chair's outlook on GDP growth, labor market strength, and external risks such as net exports and global uncertainty. Across these examples, high cosine similarity arises from both shared vocabulary and convergence in policy framing.

Together, these examples illustrate that our similarity measure captures meaningful variation in the degree to which individual FOMC participants' public statements reflect or anticipate the policy narrative presented by the Chair.

Table 1: High Cosine Similarity Examples.

Information	**Speech**	**Press Conference**
Fischer (2016/02/01) **Yellen** (2015/12/16) *Score:* 0.8152	• "...decided to raise the target range for the federal funds rate by $\frac{1}{4}$ percentage point, to $\frac{1}{4}$ to $\frac{1}{2}$ percent." • "...confidence that inflation would return to our 2 percent goal over the medium term." • "...monetary policy remains accommodative after the small increase in the federal funds."	• "...raise the target range for the federal funds rate by $\frac{1}{4}$ percentage point, bringing it to $\frac{1}{4}$ to $\frac{1}{2}$ percent." • "...reasonably confident that inflation would move back to its 2 percent objective over the medium term." • "...after today's increase, the stance of monetary policy remains accommodative."

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Table 1: High Cosine Similarity Examples. (Continued)

Information	Speech	Press Conference
Powell (2013/06/27) **Bernanke** (2013/06/19) *Score:* 0.8003	• "Inflation is currently running below the FOMC's 2 percent long-term objective." • "...purchased assets will remain on the Fed's balance sheet for some time and continue to put downward pressure on rates."	• "Inflation has been running below the Committee's longer-run objective of 2 percent." • "...adjusting the pace of asset purchases and its forward guidance regarding the target for the federal funds rate."
Fischer (2015/10/11) **Yellen** (2015/09/17) *Score:* 0.7593	• "Gross domestic product (GDP) growth in the first half of 2015 is now estimated to have been at an annual rate of 2-1/4 percent, and private forecasters are projecting GDP to continue to rise." • "Moreover, net exports have served as a significant drag on growth over the past year and recent global economic and financial developments highlight the risk that a slowdown in foreign growth might restrain U.S. economic activity." • "However, that is an expectation, not a commitment. Both the timing of the first rate increase and any subsequent adjustments to the federal funds rate target will depend critically on future developments in the economy."	• "U.S. real gross domestic product is estimated to have expanded at a 2 $\frac{1}{4}$ percent pace in the first half of the year, a notably stronger outcome than expected in June." • "Moreover, net exports were a substantial drag on GDP growth during the first half of the year, reflecting the earlier appreciation of the dollar and weaker foreign demand." • "...that the timing of the initial increase in the federal funds rate will depend on the Committee's assessment of the implications of incoming information for the economic outlook. To be clear, our decision will not hinge on any particular data release or on day-to-day movements in financial markets."

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Table 1: High Cosine Similarity Examples. (Continued)

Information	Speech	Press Conference
Brainard (2017/05/30) **Yellen** (2017/03/15) *Score:* 0.7438	• "The labor market has continued to strengthen. Payroll growth has averaged 175,000 over the past three months, more than sufficient to absorb new entrants into the labor market." • "...the underlying momentum of the domestic expansion remains solid. While U.S. consumption was weak in the first quarter of 2017, the data so far are consistent with a rebound in the current quarter. Moreover, financial conditions remain supportive of continued economic expansion despite some recent volatility."	• "Job gains averaged about 200,000 per month over the past three months, maintaining the solid pace we have seen over the past year." • "...a reassessment of the economic outlook or of the appropriate course for monetary policy. I'll have more to say about monetary policy shortly, but first I'll review recent economic developments and the outlook. The economy continues to expand at a moderate pace. Solid income gains and relatively high levels of consumer sentiment and wealth have supported household spending growth."

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Table 1: High Cosine Similarity Examples. (Continued)

Information	Speech	Press Conference
Brainard (2016/02/26) **Yellen** (2015/12/16) *Score:* 0.7466	• "The U.S. unemployment rate is now under 5 percent, compared with 10 percent at its recent peak. Even so, there is evidence that some labor market slack still remains. The United Kingdom has experienced a rapid drop in unemployment to 5.1 percent, as low as pre-crisis levels, and labor force participation has remained relatively strong." • "...many economies face common negative shocks or where negative shocks in one country are quickly transmitted across borders, it is natural to consider whether coordination can improve outcomes. Under certain conditions–such as flexible exchange rates, deep and well-regulated financial markets, and flexible product and labor markets–policies designed for the domestic economy can readily offset any spillovers from economic conditions abroad, and policies designed to address domestic conditions can achieve desirable outcomes both within the national economy and more broadly."	• "...So far this year, a total of 2.3 million jobs have been added to the economy, and over the most recent three months, job gains have averaged an estimated 218,000 per month, similar to the average pace since the beginning of the year. The unemployment rate, at 5 percent in November, is down 0.6 percentage point from the end of last year and is close to the median of FOMC participants' estimates of its longer-run normal level." • "We have considered the risks to the outlook and worried about the fact that with interest rates at zero, we have less scope to respond to negative shocks than to positive shocks that would call for a tightening of policy. That is a factor that has induced us to hold rates at zero for this long. But we recognize that policy is accommodative, and if we do not begin to slightly reduce the amount of accommodation, the odds are good that the economy would end up overshooting both our employment and inflation objectives."

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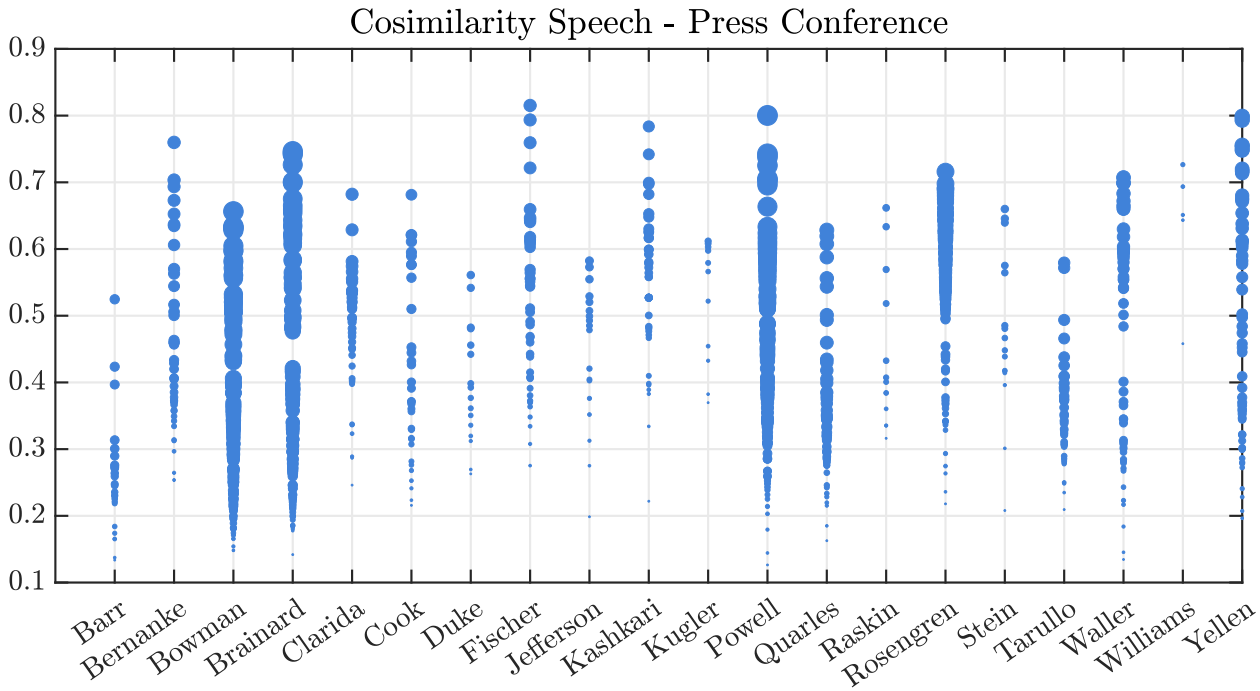
Table 1: High Cosine Similarity Examples. (Continued)

Information	Speech	Press Conference
Williams (2011/05/04) **Bernanke** (2011/04/27) *Score:* 0.7265	• "But, that doesn't mean we've regained all the ground lost during the recession. Not by a long shot. In fact, despite adding about $1\frac{1}{2}$ million jobs over the past 13 months, there are still over 7 million fewer jobs in the United States than we had before the downturn. The recovery has sputtered at times and our forward progress has been disappointingly slow." • "...Commerce Department estimated that real gross domestic product—the broadest measure of U.S. economic activity—rose just 1.8 percent on an annual basis during the first three months of this year. This is quite a bit slower than the 2.8 percent growth we saw in 2010. Unusually severe winter weather earlier this year and a few other transitory factors held down first-quarter growth."	• "...it is encouraging to see the improvement that we've seen in recent months. That being said, the pace of improvement is still quite slow, and we are digging ourselves out of a very, very deep hole. We are still something like 7 million-plus jobs below where we were before the crisis. And so, clearly, the fact that we're moving in the right direction, even though that's encouraging, doesn't mean that the labor market is in good shape." • "We haven't seen the GDP number yet. But we, like most private-sector forecasters, are expecting a relatively weak number for the first quarter, maybe something a little under 2 percent. Most of the factors that account for the slower growth in the first quarter appear to us to be transitory. They include things like, for example, lower defense spending than was anticipated, which will presumably be made up in a later quarter."

Notes: This table reports illustrative speech–press-conference pairs with high TF–IDF cosine similarity. Entries show short excerpts (ellipses denote omitted text). Scores are computed using TF–IDF representations and cosine similarity.

Figure 1 illustrates the cross-member dispersion in cosine similarity. The similarity distribution spans nearly the full possible range (0.10–0.815), indicating that individual speeches exhibit substantial heterogeneity in how closely they echo the Chair's press-conference narrative. Individual members sometimes reproduce the Chair's language almost verbatim but at other times diverge markedly in emphasis or tone. These patterns corroborate a "not a one–man show, but not egalitarian either" view of FOMC communications: Governors (and particularly the Chair) tend to cluster at higher similarity scores, consistent with more coordinated messaging, whereas regional Presidents display greater dispersion, consistent with greater idiosyncratic emphasis or policy nuance.

Figure 1: Dispersion in Speech–Press-Conference Similarity across FOMC Members



Notes: This figure plots, for each FOMC member, the distribution of cosine-similarity scores between that member's public speeches and the Chair's most recent post-meeting press conference. Each dot represents one speech–press-conference pair; dot size is proportional to the relative frequency of the score within the member's distribution. Higher values indicate greater textual alignment with the Chair's press-conference communication. The sample is restricted to speeches with an exact timestamp, consistent with the high-frequency analysis. As a result, some members (e.g., Williams) may appear with relatively few observations.

2.3 Intervening Macroeconomic News

Variation in similarity may simply reflect the macroeconomic news that arrives between the press conference and the speech. As data releases and events accumulate, the state of the economy naturally drifts away from the one described by the Chair, and a speaker commenting on the latest developments would then appear less aligned with the press conference. Under this interpretation, our interaction results in the next sections could capture the market's response to intervening macroeconomic news rather than to communication alignment per se. To verify that this is not the case, we assess whether our measure of communication alignment can be explained by the time elapsed since the preceding press conference and by intervening macroeconomic news.

For every speech, we calculate the number of days elapsed since the preceding FOMC press conference. Rather than imposing a linear relationship between elapsed time and

similarity, we approximate it with a restricted cubic spline.

$$g(d_{ism}) = \gamma_1 d_{ism} + \gamma_2 B_1(d_{ism}) + \gamma_3 B_2(d_{ism}), \tag{1}$$

with four knots at the 5th, 35th, 65th, and 95th percentiles of elapsed time. This allows alignment to evolve flexibly over the intermeeting period. d_{ism} represents the number of days elapsed between the latest press conference m and speech i by speaker s.

We then construct a database of macroeconomic releases. We exclude speeches whose delivery overlaps with a release (within a 20-minute window), leaving our sample with 905 observations. Macroeconomic surprises are standardized using the full-sample standard deviation of each indicator and aggregated into three groups: inflation news (CPI, core CPI, and PCE), labor-market news (payrolls, sign-reversed unemployment, and earnings), and activity news (GDP and industrial production). For each group, we construct both the cumulative standardized surprise, $\text{CumNews}_{ism,k}$, and the number of releases, $N_{ism,k}$. A missing surprise indicates that the release had not yet occurred and contributes to neither variable; a realized surprise of zero increases the release count but not cumulative news.

Our preferred specification is

$$Cosim_{ism} = \alpha_i + \mu_m + g(d_{ism}) + \sum_{k \in \pi, L, A} [\theta_k \text{CumNews}_{ism,k} + \phi_k N_{ism,k}] + \varepsilon_{ism}, \tag{2}$$

where α_i and μ_m denote speaker and preceding-meeting fixed effects, and k denotes the macro news category, i.e., labor market (L), inflation (π), or economic activity (A). Standard errors are two-way clustered by speaker and meeting. Table 2 presents specifications with meeting fixed effects only, speaker fixed effects only, both sets of fixed effects, and the complete set of controls.

Fixed effects explain most of the systematic variation in alignment. Meeting fixed effects alone produce an R^2 of 0.164, speaker fixed effects an R^2 of 0.201, and their joint inclusion raises the R^2 to 0.288. Conditional on meeting effects, speaker effects explain approximately 15% of the remaining variation, while meeting effects explain approximately 11% conditional on speaker effects. By contrast, adding elapsed time and macroeconomic news raises the R^2 only from 0.288 to 0.302. Alignment therefore reflects both persistent speaker characteristics and meeting-specific conditions, with speaker heterogeneity somewhat more important at the margin. Moreover, there is a large fraction of volatility that is left unexplained.

Cumulative inflation news is the only news variable with a statistically significant effect. Its estimated coefficient is -0.0092, with a two-way-clustered standard error of 0.0023. Thus, positive inflation surprises are associated with communication that is less similar to

the preceding Chair press conference. The magnitude is nevertheless limited. Given a standard deviation of cumulative inflation news of approximately 2.07, a one-standard-deviation inflation news shock reduces similarity by about 0.019, or approximately 0.12 standard deviations of the alignment index. Inflation news therefore has a statistically detectable but economically modest effect and explains little of the overall variation in alignment.

Table A1 reports the individual speaker effects relative to Barr, the omitted category. The estimates reveal substantial persistent heterogeneity: Williams, Kashkari, Kugler, and Rosengren display comparatively high alignment, whereas Tarullo, Quarles, Duke, and Bowman are closer to the omitted category. These coefficients should not be interpreted as an unconditional ranking, since they are conditional on meeting effects and the other controls. Moreover, each significance test compares one speaker with Barr rather than testing differences between pairs of speakers.

Overall, the results point to a persistent speaker-type component in communication alignment. This component may reflect stable communication styles, institutional roles, perceived authority, or market attention rather than a deliberate speech-by-speech coordination choice. This interpretation motivates examining whether monetary-policy surprises associated with more aligned speeches generate stronger financial-market responses.

2.4 Alternative Similarity Measures

To validate the cosine-similarity measure and to probe the robustness of our results, we consider two alternative methods for measuring the alignment between individual speeches and centralized messaging in FOMC press conferences.

2.4.1 LLM-based method

First, while our baseline cosine-similarity approach is transparent and widely used, it relies on a bag-of-words representation that abstracts from higher-order linguistic structure and contextual meaning and may therefore miss important variation arising from more nuanced semantic differences. To assess whether this limitation affects our results, we complement it with a more precise measure derived from a large language model (LLM)–based annotation.

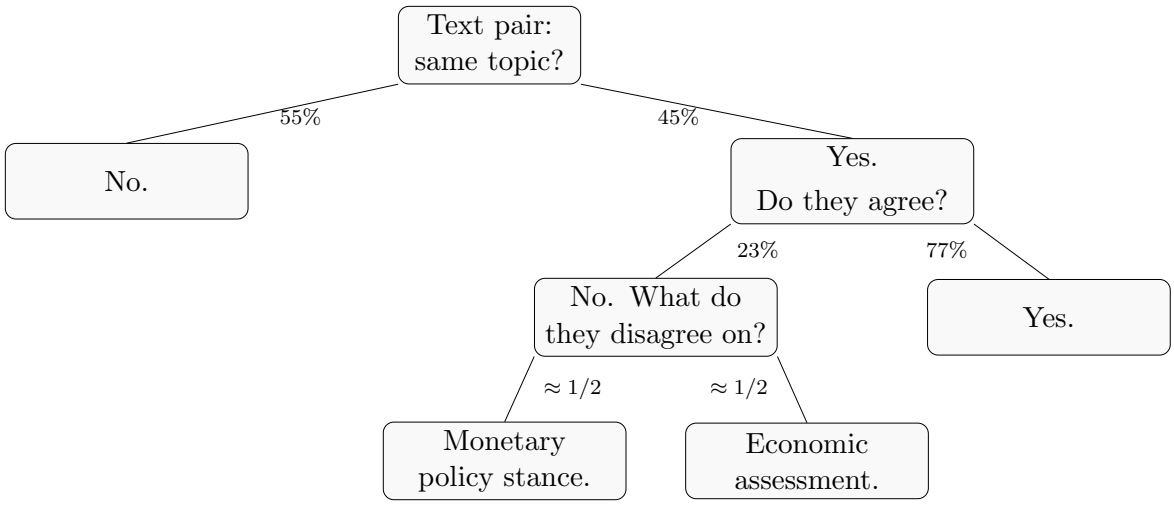
Specifically, for each speech–press-conference pair, we query ChatGPT 5.2 (Thinking) using the following prompt: "You are scoring semantic similarity between two texts. Return a continuous score between 0 and 1. Focus on meaning (topics/claims/stance), not style." We also extract a verbal explanation for the score, which we use for manual validation. This procedure yields an alternative measure of textual alignment that accounts for more complex semantic relationships, paraphrasing, and contextual cues that may not be fully captured by

Table 2: Similarity to the preceding Chair press conference

	Time FE (1)	Speaker FE (2)	Both FE (3)	Full (4)
Constant	0.4374***	0.2563***	0.2444***	0.2325***
	(0.0059)	(0.0012)	(0.0388)	(0.0528)
Days since FOMC: linear term				0.0025
				(0.0019)
Days since FOMC: spline term 1				-0.0228
				(0.0150)
Days since FOMC: spline term 2				0.0439
				(0.0261)
Cumulative inflation news				-0.0092***
				(0.0023)
Number of inflation releases				0.0046
				(0.0092)
Cumulative labor news				0.0032
				(0.0059)
Number of labor releases				-0.0004
				(0.0030)
Cumulative activity news				-0.0103
				(0.0080)
Number of activity releases				-0.0227
				(0.0201)
Meeting/time fixed effects	✓		✓	✓
Speaker fixed effects		✓	✓	✓
Observations	905	905	905	905
R^2	0.164	0.201	0.288	0.302
Adjusted R^2	0.085	0.183	0.202	0.210

Notes: The dependent variable is cosine similarity to the preceding Chair press conference. Parentheses contain standard errors two-way clustered by speaker and preceding FOMC meeting. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Figure 2: LLM-based decomposition of similarity across text pairs



word-frequency methods.

Appendix Figure A2 compares the TF–IDF cosine similarity scores to the corresponding LLM-based similarity measures across all speech–press-conference pairs in the sample. The two measures are strongly correlated, with a sample correlation of 0.76. At the same time, the dispersion around the least-squares fitted line suggests that the LLM-based measure incorporates additional dimensions of meaning beyond pure word co-occurrence.

Second, an additional advantage of the LLM-based method is that it helps interpret the source of low similarity between individual speeches and press conferences. In particular, the same framework allows us to distinguish whether low similarity reflects discussion of different topics, disagreement on a common topic, or—conditional on disagreement—differences in monetary policy stance versus assessments of economic conditions. Specifically, we use a follow-up prompt to classify (i) whether the two texts talk about the same topic; (ii) conditional on talking about the same topic, whether they express agreement or disagreement; and (iii) whether the disagreement stems from different monetary policy stances or from different assessments of economic conditions. Figure 2 summarizes this classification scheme. We find that 45% of texts talk about the same topic. Conditional on topic overlap, 77% agree, whereas 23% disagree. Conditional on disagreement, the source is split roughly evenly between monetary policy stance and economic assessment. Therefore, low similarity primarily reflects topic divergence rather than disagreement conditional on discussing the same topic. In the robustness analysis in Subsection 4, we focus on the "Yes–Yes" cases—pairs that discuss the same topic and agree—and show that conditioning on these subcells yields results very similar to those based on our high-similarity index.

2.4.2 Filtering out non-FOMC-relevant topics

The second potential weakness of our baseline measure is that it does not distinguish between topics strictly related to Fed policy and unrelated topics.

To ensure that measured similarity is topically relevant to the FOMC deliberations, we construct an alternative measure defined as the product of a speech's similarity to the contemporaneous FOMC deliberations and the Chair's press conference similarity to those same deliberations. In particular, for each meeting we compare a participant's full set of statements (using the text of FOMC meeting transcripts) to the complete set of the Chair's remarks during the press conference after FOMC meetings. We apply the same methodology to FOMC member speeches, comparing their content to the same member's remarks during the FOMC meeting. And we take the product of these two indexes. This procedure yields a participant-level topic-relevant similarity score for each meeting.

Appendix Figure A3 shows that the resulting measure is highly correlated with our simpler baseline measure, suggesting that topics other than Fed policy do not drive this variation.

3 Speeches, Content, and Market Reactions

This section examines how FOMC members' speeches affect financial markets and how those effects vary with each speech's textual similarity to the Chair's post-meeting press conference. In Subsection 3.1, we construct a high-frequency measure of monetary policy surprises around speeches and estimate their impact on interest rates and inflation expectations via local projections. In Subsection 3.2, we introduce cosine-similarity–based measures of textual proximity to the Chair's post-meeting press conference and show that market responses vary systematically with the degree of similarity.

3.1 High-Frequency Speech Surprises and Market Reactions

In this section, we document how financial markets respond to speeches by FOMC members. We first describe the construction of a high-frequency dataset capturing intraday interest-rate movements around each speech. We then use daily local projections to estimate both the statistical and economic significance of speech-induced surprises and the persistence of these effects over subsequent days.

Intraday data and event window. We match our speech similarity index to intraday changes in eurodollar (ED) futures at one- to four-quarter maturities over April 2011–Novem-

ber 2023. We retain only speeches delivered during U.S. equity-market hours on business days and exclude events in which two or more FOMC members speak simultaneously. For each qualifying speech, we measure futures changes using five-minute averages and an event window extending from ten minutes before the speech begins to twenty minutes after it ends. When unavailable, impute speech end times by adding estimated duration—calculated as the word count divided by an assumed speaking rate—to the recorded start time. After applying these restrictions and matching speeches to the futures data, our original sample of 905 speech similarity observations yields 288 usable observations.

Monetary policy speech (MPS) surprise. Within each event window, we compute intraday changes in all four maturities and extract their first principal component (PC1). We then rescale PC1 to have unit impact in the ED4 contract; this rescaled series is our *monetary policy speech* (MPS) surprise. The top panel of Appendix Figure A1 plots the MPS time series, with red markers indicating speeches by Board members, while the bottom panel displays its autocorrelation function, which shows little evidence of serial correlation.

Market Response. We assess how markets react to FOMC member speeches by running the following regression:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h MPS_t + u_{h,t} \tag{3}$$

where t represents the day when the speech occurs and $h = 1, ..., H$ is the number of days after the speech, MPS_t is the factor describing the intraday reaction of the short- to medium-term yield curve to monetary policy communication described in the previous section, and $u_{h,t}$ is an error term. Following previous work, we investigate the reaction of the yield curve and market-based measures of expected inflation. More precisely, we consider the following set of financial data: yields on U.S. Treasury securities at 1-, 5-, and 10-year constant maturities (GBY), breakeven (BE) inflation at five- and ten-year horizons, and the S&P 500.

Table 3 reports the estimated values for β_h for $h = 1, 3, 5$ when all speeches are included. These coefficients represent the impact that the speech has on financial instruments one, three, and five days after the speech, along with the portion of variance explained (R^2). A speech that increases the intraday ED4 does not have a strong impact at short maturities but does have an effect on interest rates at maturities between one and five years. Although the effect seems to be persistent and does not vanish after five days, speeches explain a tiny and diminishing portion of interest rate fluctuations over time. While not statistically significant, the point estimates of the effect on BE inflation have the expected sign (i.e., negative). The effects on the stock market, volatility, and spread are small and nonsignificant (not shown

here).

Magnitudes are small. Daily and intraday variations are expressed in percent, and the latter range between ± 0.03, i.e., 3 basis points. Thus, a 1-basis-point increase in MPS increases the five-year GBY by 1.10 basis points one day after the speech.

Table 3: Market Responses to Monetary Policy Speech (MPS) Surprises

	$h = 1$		$h = 3$		$h = 5$	
	β_h	Adj. R^2	β_h	Adj. R^2	β_h	Adj. R^2
1Y GBY	0.86***	0.11	0.71***	0.03	0.79***	0.02
5Y GBY	1.10***	0.06	0.90***	0.03	1.11***	0.02
10Y GBY	1.00***	0.05	0.83***	0.02	0.98***	0.02
10Y BE Inflation	-0.07	-0.00	-0.13	-0.00	-0.03	-0.00
5Y BE Inflation	-0.10	-0.00	-0.21	-0.00	-0.09	-0.00
SP500	0.06	0.01	-0.03	-0.00	-0.01	-0.00

Notes: Each β_h is from a separate regression of $y_{t+h} - y_{t-1}$ on the monetary policy speech (MPS) surprise on speech days. The MPS surprise is the first principal component of intraday changes in ED1–ED4 eurodollar futures in a narrow window around each speech, scaled to have a unit impact on ED4. Adj. R^2 is reported for each regression. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

3.2 High-Frequency Speech Surprises, Content, and Market Reactions

As documented in Subsection 3.1, FOMC member speeches can induce significant shifts in the yield curve but generate ambiguous responses in inflation expectations and equity prices. This pattern implies two key points. First, market participants systematically update their outlooks in response to speech content. Second, the direction of these updates–whether the speech is interpreted as positive or negative news–cannot be inferred from yield-curve shifts alone. Rather than reflecting a lack of informational value, this interpretative ambiguity is likely a natural consequence of the heterogeneous nature of speeches. Some speeches emphasize projections for real economic activity, while others convey information about the stance and future path of monetary policy. This motivates asking whether one observable dimension of this heterogeneity, which is the degree to which a speech aligns with the Chair's preceding message, systematically influences its transmission.

Empirical work has shown that these different information types can produce similar yield-curve responses but divergent effects on inflation expectations and asset prices (e.g., Andrade and Ferroni, 2021). Moreover, variation in focus on short-run versus long-run dynamics, as well as differences in individual FOMC participants' perspectives, further contribute to the multiplicity of market interpretations.

Speech content is inherently multidimensional, making it difficult to construct concise summary metrics. We therefore focus on one interpretable margin: the degree of textual similarity between an individual speech and the Chair's preceding post-meeting press conference. This allows us to ask whether financial markets respond differently when a speech reinforces the Chair's earlier message rather than conveying distinct information, and in particular whether aligned speeches are redundant or instead provide incremental information that strengthens monetary policy transmission. Anchoring our analysis to the press conference–a benchmark widely studied in the literature on monetary policy communication and its effects on financial markets and macroeconomic outcomes (e.g., Bauer and Swanson, 2023a,b)—facilitates direct comparison with existing work.

We organize the analysis in three steps. First, we introduce the text-based measure of each speech's similarity to the Chair's post-meeting press conference. Second, we examine how the correlation between monetary policy speech surprises and daily changes in key financial variables varies across different levels of similarity. Finally, we estimate dynamic projections in which speech surprises are interacted with similarity measures to quantify how the content of a speech conditions its effect on yields, inflation expectations, and other asset prices over time.

In the subsequent analysis, we classify each speech into one of three categories based on its similarity score, ℓ_t: (i) *Far*: $\ell_t < \mu(\ell) - \sigma(\ell)$, (ii) *Neutral*: $\mu(\ell) - \sigma(\ell) \leq \ell_t \leq \mu(\ell) + \sigma(\ell)$, and (iii) *Close*: $\ell_t > \mu(\ell) + \sigma(\ell)$, where $\mu(\ell)$ and $\sigma(\ell)$ denote the sample mean and standard deviation of $\{\ell_t\}$, respectively.

We then compute the correlation between the monetary policy surprises for speeches in each of the three categories and the daily changes in yields on U.S. Treasury securities at 1-, 5-, and 10-year constant maturities (GBY) and breakeven (BE) inflation at five- and ten-year horizons. Table 4 reports the correlations between financial variables and the monetary policy speech surprise conditional on its similarity to the press conference.

Consistent with the findings of Subsection 3.1, monetary policy speech surprises are typically correlated with daily shifts in the yield curve but exhibit little association with market-based inflation expectations (see the "All" column in Table 4). When we partition speeches based on their similarity score, distinct patterns emerge. Speeches with low cosine similarity are associated with positive co-movement between nominal yields and breakeven inflation—especially at longer maturities—whereas speeches that exhibit a high cosine similarity produce yield increases that coincide with declines in inflation expectations. The one-year yield response is particularly pronounced for high cosine similarity speeches. High cosine similarity speeches consistently yield negative correlations between nominal interest rates and inflation expectations.

Table 4: Correlations between Financial Variables and MPS Surprises by Similarity Bin

	$h = 1$ (one day after)				$h = 3$ (three days after)				$h = 5$ (five days after)			
	All	Far	Neutral	Close	All	Far	Neutral	Close	All	Far	Neutral	Close
1Y GBY	0.33	0.21	0.12	0.68	0.19	0.23	0.11	0.23	0.16	0.21	0.04	0.26
5Y GBY	0.26	0.26	0.16	0.45	0.17	0.23	0.16	0.17	0.16	0.17	0.08	0.28
10Y GBY	0.23	0.21	0.15	0.41	0.16	0.20	0.13	0.18	0.15	0.15	0.09	0.25
10Y BE Inflation	-0.03	0.08	-0.03	-0.12	-0.04	0.17	-0.02	-0.20	-0.01	0.10	-0.02	-0.06
5Y BE Inflation	-0.03	0.13	-0.02	-0.19	-0.05	0.21	-0.02	-0.29	-0.02	0.13	-0.01	-0.14
SP500	0.10	0.35	0.03	0.04	-0.03	0.01	0.02	-0.13	-0.01	0.15	-0.01	-0.14

Notes: Entries are sample correlations between the monetary policy speech (MPS) surprise on day t and daily changes in each financial variable measured $h \in \{1, 3, 5\}$ days after the speech. "All" uses all speeches; "Far," "Neutral," and "Close" split speeches into groups based on the speech–press-conference similarity index ℓ_t: Far if $\ell_t < \mu(\ell) - \sigma(\ell)$, Close if $\ell_t > \mu(\ell) + \sigma(\ell)$, and Neutral otherwise. The sample sizes for Far, Neutral, and Close are 53, 185, and 50, respectively.

Market Response. To assess how variation in speech content relative to the Chair's message influences the market impact of speech-induced surprises, we estimate the following interaction specification.

$$y_{t+h} - y_{t-1} \;=\; \alpha_h \;+\; \beta_h \,\mathrm{MPS}_t \;+\; \gamma_h \,\ell_t \;+\; \delta_h \left(\mathrm{MPS}_t \times \ell_t\right) \;+\; u_{h,t}, \tag{4}$$

for horizons $h = 0, 1, \ldots, H$, where ℓ_t is our baseline speech–press-conference cosine similarity. By allowing ℓ_t to vary continuously in $[0, 1]$, we trace how the marginal effect of the speech surprise,

$$\frac{\partial}{\partial\, \mathrm{MPS}_t}\left(y_{t+h} - y_{t-1}\right) \;=\; \beta_h \;+\; \delta_h \,\ell_t, \tag{5}$$

evolves with similarity to the Chair's message. Confidence bands for this varying impact are constructed so that

$$Var\left(\beta_h + \delta_h \,\ell_t\right) = Var(\beta_h) + \ell_t^2 Var(\delta_h) + 2\,\ell_t Cov(\beta_h, \delta_h).$$

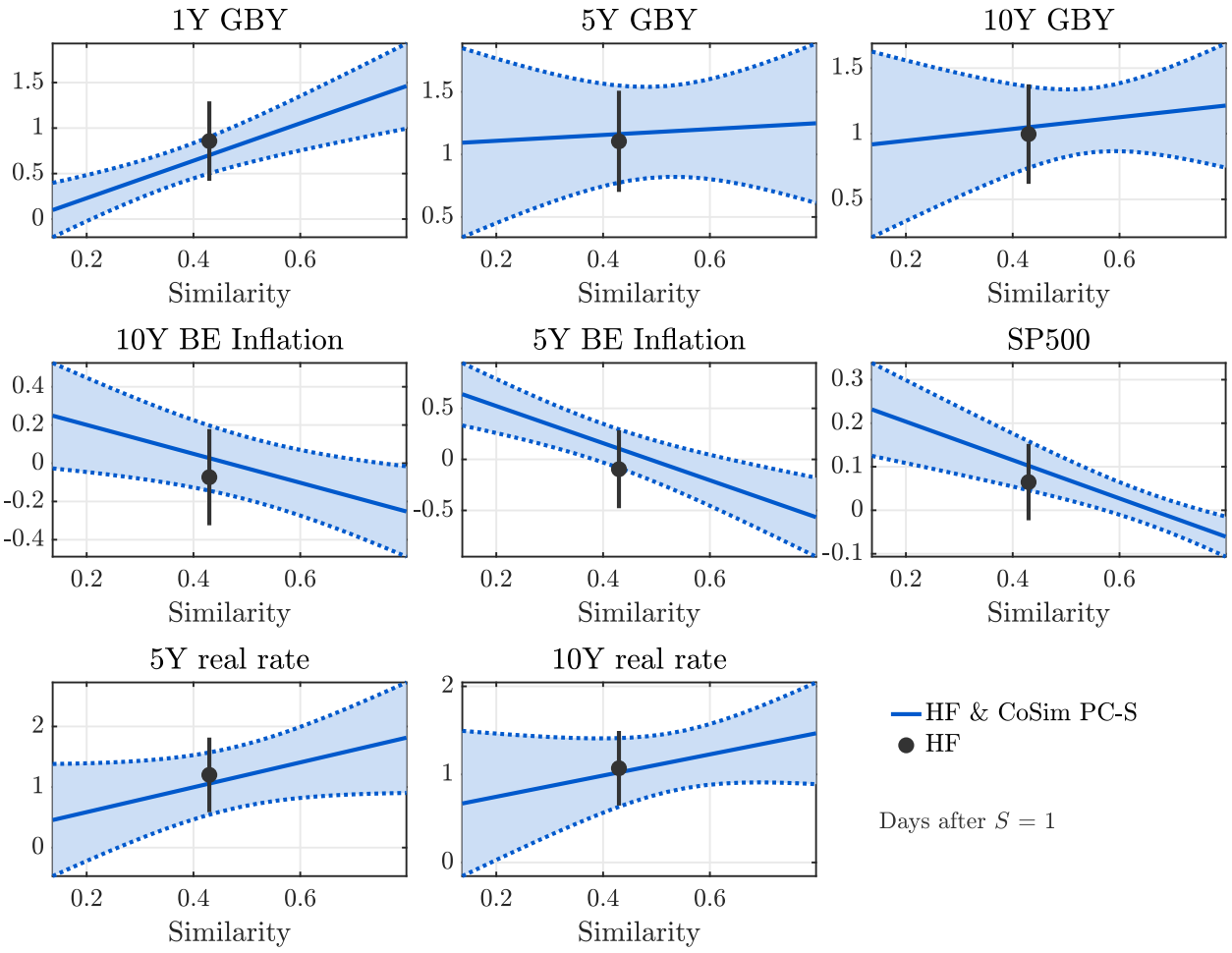
Figure 3 displays the financial market responses one day after a monetary policy speech, using our baseline measure of similarity. Each panel plots the effect of a tightening monetary policy surprise as a function of similarity between the speech and the Chair's post-meeting message: values on the left correspond to speeches that exhibit a low cosine similarity, while those on the right are closer to the press conference. The black circle denotes the unconditional (average) effect across all observations.

The results yield two main insights. First, when the speech content exhibits a high cosine similarity (toward the right of the plots), short-to-intermediate interest rates (six months to one year) exhibit a more pronounced response, while inflation expectations decline—unlike

in other regions where they are typically insignificant or mildly positive. In contrast, the long end of the yield curve shows weaker and statistically insignificant movements. This pattern is broadly consistent with a temporary monetary policy tightening, where markets expect near-term adjustments without implications for the long-run stance.

Second, when the speech content exhibits a low cosine similarity (toward the left), the response of the term structure steepens: interest rates at longer maturities react more strongly than short-term rates. This is accompanied by rising inflation expectations and a positive stock market response. These patterns can be interpreted in two complementary ways. One possibility is that the speech conveys additional information about future economic fundamentals (Delphic).

Figure 3: Responses to MPS Surprises across Speech–Press-Conference Similarity



Notes: Each panel reports the estimated marginal effect, as captured by equation (5), of a tightening monetary policy speech (MPS) surprise on the indicated financial variable at horizon $h = 1$, as a function of the speech–press-conference similarity index $\ell_t \in [0, 1]$. The MPS surprise is measured in a narrow intraday window around the speech and normalized so that a one basis point shock raises the one-year government-bond yield by 1 basis point within that window. The solid blue line plots the conditional marginal effect $\beta_h + \delta_n \ell_t$ from equation (5). The vertical axis is measured in basis points for yields, breakeven inflation rates, and real rates, and in percent for the S&P 500. Shaded areas and dotted lines denote pointwise confidence bands. The black circle reports the unconditional marginal effect from regression (3) without the similarity interaction and is plotted at the sample mean of ℓ_t for comparison; vertical bars report its confidence interval.

Daily Local Projections. To isolate financial market responses to monetary policy speeches that closely resemble the Chair's message, we restrict the sample to episodes in which the speech content exhibits a high degree of textual similarity to the post-meeting press conference. Specifically, we retain only those speeches for which the cosine similarity index exceeds one standard deviation above the mean. For this restricted sample, we estimate the following regression:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h MPS_t^{\star} + u_{h,t},$$

for $h = 0, 1, \ldots$, where $MPS_t^{\star}$ captures the high-frequency variation in interest rates associated with speeches classified as highly similar to the Chair's press conference.
Figure 4 plots the responses of financial variables over time. The black line shows the baseline response to all monetary policy speeches, while the blue line isolates the response to speeches with high textual similarity. The results are consistent with the baseline findings: tightening speeches that resemble the Chair's message lead to a significant increase in interest rates at maturities between six months and one year, a persistent decline in inflation expectations, and little to no effect at the long end of the yield curve.

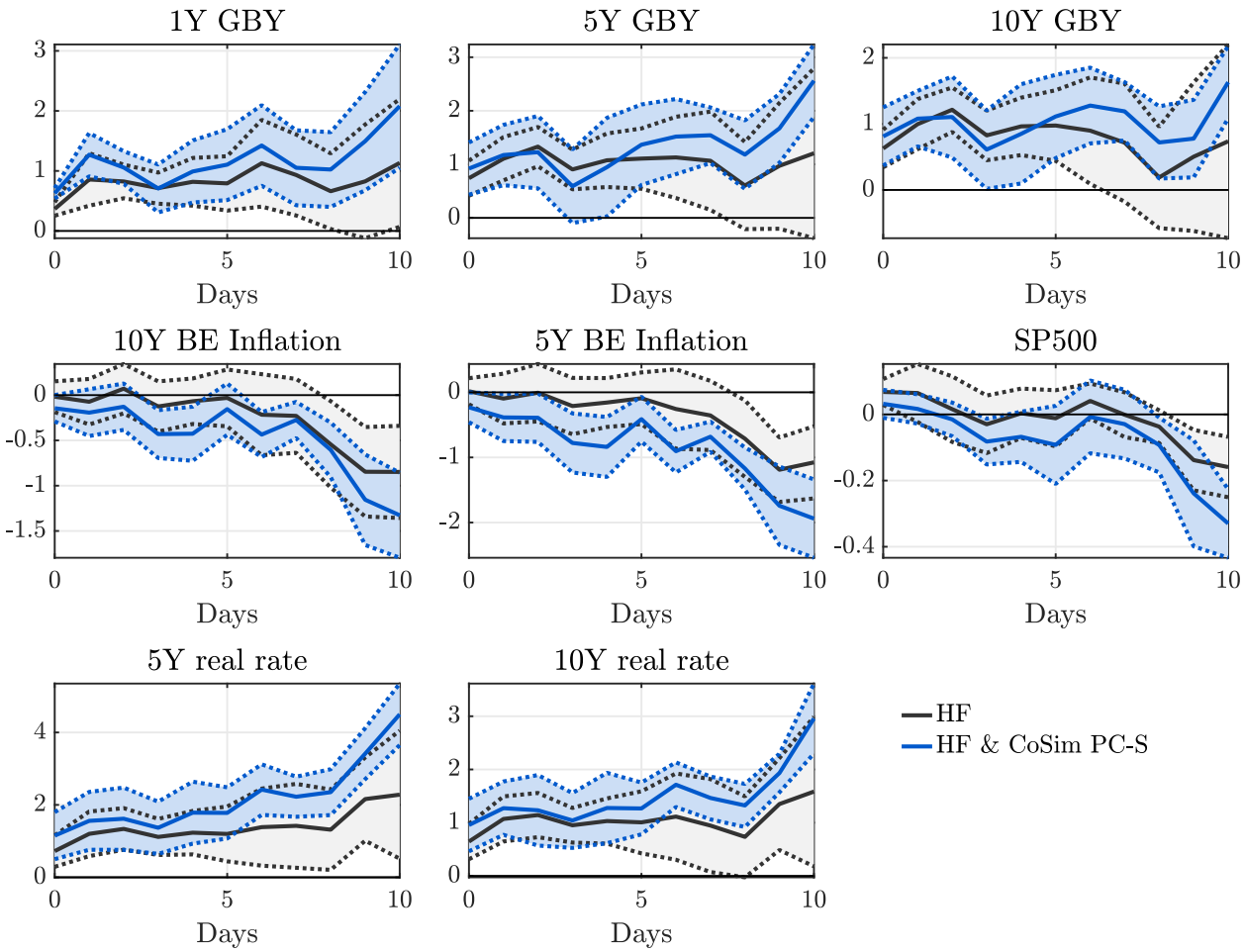
4 Robustness

We assess robustness along eight dimensions: (i) varying the event-study horizon; (ii) excluding speeches delivered by the FOMC Chair and those delivered by FOMC members without voting rights; (iii) replacing TF–IDF cosine similarity with an LLM-based semantic similarity score; (iv) restricting attention to text pairs classified by the LLM as discussing the same topic and agreeing; (v) using an alternative similarity index that filters out non–FOMC-relevant topics by conditioning similarity on contemporaneous FOMC deliberations; (vi) excluding the Q&A portion of the Chair's press conferences and recomputing the speech–press-conference similarity measures; (vii) orthogonalizing our measure of communication alignment with respect to the macroeconomic news released between the press conference and the speech; (viii) controlling for the following factors (standalone or jointly): the ELB period, the time elapsed between events, the regional Fed Presidents, and large events.

4.1 Alternative Time Horizons

Our baseline results evaluate the interaction specification at horizon h=1, i.e., using daily changes measured one day after each speech. Appendix Figures A4 and A5 replicate Fig-

Figure 4: Dynamic Responses to MPS Surprises: High-Similarity Speeches vs. Full Sample



Notes: This figure reports daily local-projection impulse responses of the indicated financial variables to an MPS tightening shock normalized to raise the one-year government-bond yield by 1 basis point within the intraday speech window. Responses are measured using daily changes from the day before the speech to each horizon after the speech. The vertical axis is in basis points for yields, breakeven inflation rates, and real rates, and in percent for the S&P 500. Because the shock is measured intraday while the responses are measured at daily frequency, the day-0 response of the one-year government-bond yield is not mechanically equal to 1 basis point. The black line shows estimates using all speeches in the sample. The blue line restricts the sample to speeches with high textual similarity to the Chair's most recent post-meeting press conference (similarity exceeding one standard deviation above the mean). Shaded areas denote pointwise confidence bands. The horizontal axis reports days after the speech.

ure 3 for h=0 (same-day changes) and h=2 (two-day changes). The qualitative pattern is unchanged. When a speech is more closely aligned with the Chair's post-meeting press conference, a tightening MPS surprise primarily affects short-to-intermediate maturities and is associated with lower breakeven inflation. When similarity is low, the response tilts toward longer maturities and inflation expectations do not decline. Overall, the similarity-conditioned heterogeneity in market responses is not sensitive to the choice of horizon.

4.2 Excluding FOMC Chair and Nonvoter Speeches

To probe the robustness of our baseline results, we first verify that they are not driven by speeches delivered by the FOMC Chair outside official FOMC events. Figures A6 and A7 report the resulting estimates and are the direct analogues of Figures 3 and 4 in the main text, respectively. As demonstrated by these figures, our results are hardly affected once we remove Chair speeches. This suggests that markets extract clues about monetary policy not only from the Chair's speeches but also from the committee's many voices.

Moreover, our sample mostly contains speeches by FOMC members with voting rights at the time of the speech, and only a few are delivered by "nonvoters." Removing them does not affect the conclusions obtained using the full sample; see Figures A8 and A9.

4.3 Alternative Similarity Measure: LLM-Based Similarity

While transparent, our simple bag-of-words approach of measuring similarity between individual speeches and press conferences may miss semantic proximity that arises through paraphrasing, contextual framing, or differences in wording. For robustness, we repeat the analysis in Subsection 3.2 using an LLM-based semantic similarity score in place of cosine similarity.

Specifically, we re-estimate equation (4) replacing the cosine-similarity index ℓ_t with its LLM-based counterpart $\tilde{\ell}_t \in [0, 1]$, and trace the implied marginal effect of a monetary policy speech surprise, $\beta_h + \delta_h \tilde{\ell}_t$, across values of semantic similarity.

Figure A10 (LLM analogue of Figure 3) delivers the same qualitative message as the cosine-similarity results. When speeches are semantically close to the Chair's message, the tightening surprise produces larger increases at short-to-intermediate maturities (roughly six months to one year) and a decline in market-based inflation expectations, with comparatively muted movements at the long end. When speeches are semantically distant, the term-structure response tilts toward longer maturities and inflation expectations do not fall.

We also replicate the dynamic exercise underlying Figure 4. Mirroring the baseline cutoff, we classify a speech as "high similarity" when its LLM similarity exceeds one standard

deviation above the mean and estimate daily local projections on this restricted sample. Figure A11 (LLM analogue of Figure 4) again closely matches the baseline: semantically aligned tightening speeches generate a more pronounced and persistent response in short-to-intermediate rates and real rates, alongside a sustained decline in breakeven inflation, while longer-maturity yields move little. Overall, the near-coincidence between the cosine- and LLM-based results indicates that our findings are robust to an alternative notion of similarity that directly targets semantic content.

4.4 LLM-Based Topic Agreement

We also examine whether our results continue to hold when we restrict attention to the subset of speech–press-conference pairs that, according to the LLM classification summarized in Figure 2, discuss the same topic and agree. This restriction isolates the "Yes–Yes" subcells.[2] Appendix Figure A14 shows that conditioning on these subcells yields dynamic responses very similar to those obtained using our high-similarity index. This finding further supports the interpretation that our baseline results capture meaningful alignment in policy communication.

4.5 Alternative Similarity: Filtering Out Non-FOMC-Relevant Topics

We verify that our results capture similarity in FOMC-relevant content rather than unrelated topics. To do so, we replicate the baseline analysis using an alternative similarity measure. Instead of the cosine similarity between each speech and each press conference, we compute (i) the cosine similarity between the speech and FOMC deliberation text and (ii) the cosine similarity between the press conference and the same deliberation text, and use the product of these two similarities as our measure. The results (Appendix Figures A12 and A13) are consistent with those obtained using the baseline measure.[3]

4.6 Excluding the Press Conference Q&A

Our baseline similarity measure is computed using the Chair's full post-meeting press conference, including both the prepared opening statement and the subsequent question-and-answer session with journalists. As a robustness check, we recompute the speech–press-

[2]Rather than a continuous measure, this variable is an indicator equal to one for text pairs in that category.

[3]The difference is that these results exhibit wider confidence intervals, which likely reflects the much shorter sample period available for that specification.

conference similarity measure after excluding the Q&A portion and retaining only the Chair's prepared remarks.

Figures A15 and A16 report the counterparts of Figures 3 and 4 using this alternative similarity measure. The results are virtually unchanged. Speeches that are more closely aligned with the Chair's prepared remarks continue to generate stronger responses at short-to-intermediate maturities together with declines in inflation expectations, whereas less aligned speeches are associated with relatively stronger movements at the long end of the yield curve and weaker disinflationary effects. Likewise, the dynamic local-projection responses remain quantitatively and qualitatively similar to the baseline specification. Overall, our findings are therefore not driven by the inclusion of the Q&A portion of the press conference when constructing the similarity measure.

4.7 Removing Macroeconomic News

We orthogonalize our baseline cosine-similarity measure with respect to the macroeconomic news released between the press conference and the speech. To this end, we regress the cosine-similarity score on the macroeconomic-news indexes described in Subsection 2.3 and denote the resulting least-squares residual by $\ell_t^{\perp}$. We interpret $\ell_t^{\perp}$ as a macro-news-orthogonal measure of communication alignment.[4] The results (Appendix Figures A17 and A18) are consistent with those obtained using the baseline measure.

4.8 Controlling for various factors

As a last robustness exercise, we consider a number of confounding factors in our baseline specification, i.e. equation (4). In particular, we specify the following regression:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h \, \text{MPS}_t + \gamma_h \, \ell_t + \delta_h \left(\text{MPS}_t \times \ell_t\right) + \Gamma_h' \, Z_t + \Delta_h' \left(\text{MPS}_t \times Z_t\right) + e_{h,t}, \quad (6)$$

where Z_t is an $n_z \times 1$ vector containing the control variables, and Γ_h and Δ_h are column vectors of appropriate dimension. We consider the following factors in Z_t, separately or simultaneously (All): a dummy variable equal to one during the ELB period (ELB), the time elapsed since the preceding press conference (Time Elapsed), a dummy variable equal to one when the speaker is a Regional Fed President (Regional Pres), and a variable capturing extreme events ($MPS \times |MPS|$). Finally, to verify that our results are not driven by outliers, we winsorize the MPS distribution at the 1st and 99th percentiles. Table 5 displays

[4] The construction of the macroeconomic news database and the relevant indexes is discussed in Section 2.3.

the estimates of the coefficient on the interaction between similarity and MPS, δ_h, in the baseline specification in equation (4) and in the specification with various controls in equation (6). These confounders do not seem to influence our main results.

Table 5: Estimates of the coefficient on the interaction between similarity and MPS with and without controls

	Baseline	ELB	Time Elapsed	Regional Pres	$MPS \times \lvert MPS \rvert$	All	MPS_w
1Y GBY	2.06***	2.06***	2.14***	2.03***	1.88***	1.96***	2.61*
5Y GBY	0.23	0.23	0.36	0.22	0.46	0.49	0.98
10Y GBY	0.45	0.44	0.48	0.43	0.68	0.63	0.92
10Y BE Inflation	-0.76**	-0.76**	-0.93**	-0.78**	-0.84***	-1.06***	-1.59***
5Y BE Inflation	-1.82***	-1.83***	-1.91***	-1.87***	-1.88***	-2.07***	-3.35***
SP500	-0.44***	-0.44***	-0.47***	-0.44***	-0.44***	-0.48***	-0.59*

Notes: Each entry reports the estimate of δ_1 in equation (6) from a separate regression of $y_{t+1} - y_{t-1}$ on the monetary policy speech (MPS) surprise on speech days, cosine similarity, and various controls. The MPS surprise is the first principal component of intraday changes in ED1–ED4 eurodollar futures in a narrow window around each speech, scaled to have a unit impact on ED4. The first column reports the baseline estimates; the following columns add one control at a time: a dummy variable equal to one during the ELB period (ELB), the time elapsed since the preceding press conference (Time Elapsed), a dummy variable equal to one when the speaker is a regional Fed President (Regional Pres), and a variable capturing large events ($MPS \times |MPS|$). The last column reports the coefficient estimate obtained after winsorizing the MPS distribution at the 1st and 99th percentiles (MPS_w). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5 An NK Model with Speech-Driven Expectations

We develop a New Keynesian model in which the Chair's communication and a member's speech provide public information about future monetary policy. The model separates two mechanisms. In the baseline, similarity measures the precision of a common policy message: more similar communication strengthens the response to news of future tightening. This mechanism generates a negative relationship between similarity and the inflation response to fixed observed messages. We then allow the member's speech to contain information about future demand. This extension allows positive communication news to raise both expected interest rates and inflation when similarity is low.

5.1 Economic Environment

Time is quarterly. A representative household chooses consumption C_t, labor N_t, and nominal bonds B_t to maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma} - 1}{1-\sigma} - \chi \frac{N_t^{1+\varphi}}{1+\varphi} \right] \tag{7}$$

subject to its nominal budget constraint,

$$P_t C_t + B_t = W_t N_t + R_{t-1} B_{t-1} + P_t D_t, \tag{8}$$

where P_t is the price level, W_t the nominal wage, D_t real dividends, and R_t the gross return on a one-period bond purchased at t. Let $\Pi_t = P_t/P_{t-1}$ and $w_t = W_t/P_t$. Optimality requires

$$\chi N_t^{\varphi} = C_t^{-\sigma} w_t, \qquad C_t^{-\sigma} = \beta R_t \mathbb{E}_t \left[\frac{C_{t+1}^{-\sigma}}{\Pi_{t+1}} \right]. \tag{9}$$

Expectations are rational and conditional on the public information specified below. Bonds are in zero net supply.

A competitive final-good sector aggregates differentiated goods with elasticity of substitution $\epsilon_p > 1$. Intermediate firm j faces demand $Y_t(j) = [P_t(j)/P_t]^{-\epsilon_p} Y_t$ and produces with linear technology, $Y_t(j) = N_t(j)$. Changing its price entails a resource cost $\phi_p[P_t(j)/P_{t-1}(j) - 1]^2 Y_t/2$. Firms maximize the expected present value of profits using the household's real stochastic discount factor, $M_{t,t+1} = \beta(C_{t+1}/C_t)^{-\sigma}$. Symmetric price setting implies

$$\phi_p(\Pi_t - 1)\Pi_t = 1 - \epsilon_p + \epsilon_p w_t + \mathbb{E}_t \left[M_{t,t+1} \phi_p (\Pi_{t+1} - 1)\Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right]. \tag{10}$$

Labor and goods markets clear when

$$N_t = Y_t, \qquad C_t = Y_t \left[1 - \frac{\phi_p}{2} (\Pi_t - 1)^2 \right]. \tag{11}$$

Monetary policy follows a Taylor rule without interest-rate smoothing:

$$\log(R_t/\overline{R}) = \phi_\pi \log \Pi_t + \phi_y \log(Y_t/\overline{Y}) + \theta_t, \tag{12}$$

$$\theta_{t+1} = \rho_\theta \theta_t + \sigma_\theta \varepsilon_{t+1}^{\theta}, \qquad \varepsilon_{t+1}^{\theta} \overset{\text{iid}}{\sim} \mathcal{N}(0,1). \tag{13}$$

The deterministic steady-state allocation has $\overline{\Pi} = 1$, $\overline{R} = \beta^{-1}$, and $\overline{C} = \overline{Y} = \overline{N} = 1$, with $\chi = (\epsilon_p - 1)/\epsilon_p$.

5.2 Communication, Similarity, and Beliefs

We interpret greater textual similarity as indicating that the two communications more precisely convey a common underlying policy message. Importantly, an aligned speech is not a repetition of the Chair's signal: it provides an additional noisy observation of the same future policy innovation. This captures the idea that subsequent aligned speeches can

contain incremental information even when they reinforce the Chair's message.

Before making decisions at date t, agents observe the current policy state θ_t, similarity ζ_t, and two signals about the future monetary-policy innovation $\varepsilon^{\theta}_{t+1}$: the Chair's signal, s^C_t, and the FOMC member's speech signal, s_t.

$$s^C_t = \varepsilon^{\theta}_{t+1} + \nu_B(\zeta_t)\eta^C_t, \qquad s_t = \varepsilon^{\theta}_{t+1} + \nu_B(\zeta_t)\eta_t, \qquad \nu^2_B(\zeta) = \frac{1-\zeta}{\zeta}. \tag{14}$$

The measurement errors η^C_t and η_t are independent standard normal variables; they are also independent over time and of the policy innovations and similarity. Agents observe neither the future innovation nor either error separately. Thus, the speech provides a second measurement of policy news without repeating the Chair's measurement error. The model abstracts from the within-period timing of the press conference and the subsequent speech; both messages belong to the information set when decisions are made.

We map the observed cosine similarity into the correlation of the two signals conditional on the communication environment:

$$\mathrm{Corr}(s^C_t, s_t \mid \zeta_t = \zeta) = \frac{1}{1+\nu^2_B(\zeta)} = \zeta.$$

We interpret this equality as a measurement assumption linking text similarity to the model's information structure. Similarity is an exogenous draw, $\zeta_t \overset{\text{iid}}{\sim} F_\zeta = \mathrm{Beta}(a_\zeta, b_\zeta)$, and is independent of the innovations. Equal noise variances give the Chair's message and the speech symmetric precision in the baseline.

Bayesian updating yields

$$\varepsilon^{\theta}_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(\mu_t, V_B(\zeta_t)), \qquad \mu_t = \frac{\zeta_t}{1+\zeta_t}(s^C_t + s_t), \qquad V_B(\zeta_t) = \frac{1-\zeta_t}{1+\zeta_t}, \tag{15}$$

where $\mathcal{F}_t$ denotes public information after the messages arrive. For fixed positive messages, higher similarity raises the posterior mean of the policy innovation and lowers its posterior variance. Greater precision gives the same observed messages more weight in agents' posterior beliefs. At $\zeta = 0$, the uninformative limit restores the prior; at $\zeta = 1$, consistent messages perfectly reveal the innovation. The Beta distribution assigns no mass to either endpoint.

Equilibrium. The recursive state is $X_t = (\theta_t, \zeta_t, \mu_t)$. A symmetric recursive competitive equilibrium consists of policy functions $\{\mathcal{C}, \mathcal{N}, \mathcal{Y}, \mathcal{P}, \mathcal{R}, \mathcal{W}\}(X)$ for consumption, labor, output, gross inflation, the gross nominal interest rate, and the real wage. At every admissible

Table 6: Parameter values

Parameter	Value	Interpretation or target
β	0.99	Quarterly discount factor
σ, φ	$2, 1$	Consumption curvature and inverse Frisch elasticity
ϵ_p	6	Elasticity of substitution across goods
χ	5/6	Unit reference consumption and labor
ϕ_π, ϕ_y	$1.5, 0.3$	Taylor-rule responses to inflation and output
κ	0.0429	Local Phillips-curve slope on log output
ϕ_p	349.65	Implied Rotemberg adjustment-cost coefficient
$\rho_\theta, \sigma_\theta$	$0.10, 0.0025$	Persistence and volatility of the monetary policy shocks
$\overline{\zeta}, \mathrm{sd}(\zeta)$	$0.4328, 0.1494$	Empirical cosine-similarity moments
a_ζ, b_ζ	$4.3245, 5.6674$	Beta parameters implied by those moments

state, these functions satisfy the household conditions (9), the firm pricing condition (10), the monetary policy rule (12), and market clearing (11). Bond holdings are zero, and firm dividends satisfy $D_t = C_t - w_t N_t$, ensuring the household budget constraint (8). The usual no-Ponzi and transversality conditions apply.

Conditional expectations are taken under the transition kernel $Q(dX' \mid X)$ implied by the policy process (13), the signal structure (14), and Bayesian updating (15). In particular,

$$\theta_{t+1} \mid X_t \sim \mathcal{N}\big(\rho_\theta \theta_t + \sigma_\theta \mu_t,\, \sigma_\theta^2 V_B(\zeta_t)\big)\,.$$

Next-period similarity is drawn independently from F_ζ, and $\mu_{t+1} \mid \zeta_{t+1}$ is normal with mean zero and variance $1 - V_B(\zeta_{t+1})$, independently of θ_{t+1} conditional on X_t and ζ_{t+1}. Thus, posterior variance is a known function of similarity and requires no additional state variable. We use a global solution method to evaluate the nonlinear equilibrium under the similarity-dependent posterior distributions. We solve the model by time iteration with Chebyshev polynomial projection, evaluating conditional expectations by numerical quadrature under the state-specific posterior distribution; see Appendix B for details.

5.3 Calibration and the Similarity Profile

Table 6 reports the common calibration. The Rotemberg coefficient is set to $\phi_p = (\epsilon_p - 1)(\sigma + \varphi)/\kappa = 349.65$ so that the local Phillips-curve slope on log output is $\kappa = 0.0429$. The policy innovation has standard deviation $\sigma_\theta = 0.0025$, equivalent to 25 basis points in the log policy-rule disturbance. The Beta distribution matches the empirical mean and standard deviation of speech–press-conference cosine similarity, 0.4328 and 0.1494. At mean similarity, $V_B(\overline{\zeta}) = 0.3959$.

To compare the model with the empirical finding in Figure 3, Figure 5 reports model-implied response profiles across similarity. We evaluate the nonlinear equilibrium at $\theta_t = 0$, the deterministic steady-state value of the policy disturbance. For each $\zeta \in [0, 1]$, we compare a positive communication episode, $s_t^C = s_t = 1$, with a zero-message benchmark, $s_t^C = s_t = 0$, evaluated at the same value of similarity.

In the baseline model, the positive messages imply a posterior mean $\mu_t = 2\zeta_t/(1+\zeta_t)$ for the future monetary-policy innovation, whereas the zero messages imply $\mu_t = 0$. Because the two states are evaluated at the same value of ζ_t, they also have the same posterior variance as per equation (15).

Let $i_t = \log R_t$ and $\pi_t = \log \Pi_t$. For each outcome $j \in \{\mathbb{E}_t i_{t+1}, \pi_t, \mathbb{E}_t \pi_{t+1}\}$, we compute

$$r_j(\zeta) = 10^4 \left[y_j(s_t^C = 1, s_t = 1; \zeta) - y_j(s_t^C = 0, s_t = 0; \zeta) \right],$$

where the outcomes are evaluated using the nonlinear equilibrium policy functions and model-consistent conditional expectations. We normalize each response by the increase in the expected next-quarter short rate generated by the same communication experiment at mean similarity:

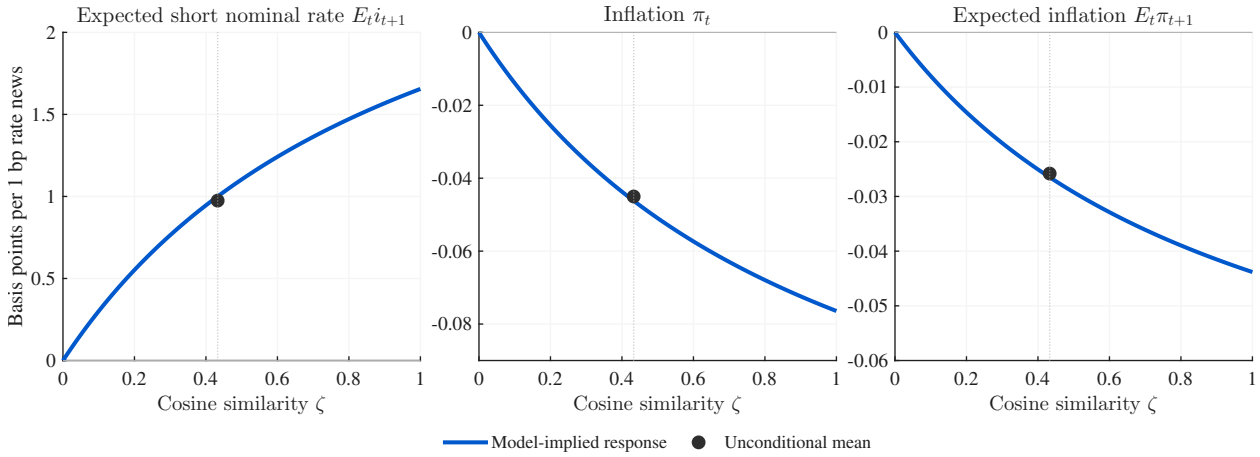
$$\widetilde{r}_j(\zeta) = \frac{r_j(\zeta)}{r_{Ei}(\bar{\zeta})}. \tag{16}$$

All profiles share this denominator, so the expected-rate response equals one at mean similarity and varies with similarity.

The three panels of Figure 5 report, from left to right, the responses of the expected short rate $\mathbb{E}_t i_{t+1}$, current inflation π_t, and expected inflation $\mathbb{E}_t \pi_{t+1}$. Each panel shows the nonlinear response profile defined in equation (16). See Appendix B for details. As shown in the figure, higher similarity amplifies the increase in the expected short rate and the decline in current and expected inflation, qualitatively consistent with the empirical pattern in Figure 3. Intuitively, as similarity rises, the same positive messages imply a larger posterior mean of the future policy innovation and lower posterior uncertainty, making the anticipated tightening both stronger and more precisely identified.

The baseline illustrates how greater precision strengthens the response to the same observed messages. To also generate the sign reversal observed in the data at low similarity, we allow speeches to convey news about future demand, with the demand-news component becoming stronger as similarity declines.

Figure 5: Similarity and the Response to Policy News



Notes: The baseline responses (blue solid curves) compare unit positive Chair and speech signals with zero signals at the same similarity and zero current policy state. Black circles are the similarity-averaged responses, plotted at $\overline{\zeta} = 0.4328$. Responses are divided by the expected short-rate response at mean similarity (12.5152 basis points). The plotted objects are in basis points of the outcome per basis point of reference rate news and are not annualized.

5.4 An Extension with News about Future Demand

The baseline model explains why beliefs respond more strongly to the same messages as communication becomes more aligned, but it cannot account for the positive inflation response observed at low similarity. We therefore allow a speech that departs from the Chair's message to contain information about future demand. As similarity falls, the speech becomes relatively less informative about the common policy message and relatively more informative about this additional fundamental.

We now allow the member's speech to contain information about a second fundamental, a preference shifter r_t^* that raises desired current spending relative to future spending. All parameters in Table 6 are retained. Two changes are required: an additional demand state and an additional component of the speech signal.

First, multiply the period utility in equation (7) by a positive weight Ξ_t, with $\Xi_0 = 1$ and

$$\frac{\Xi_{t+1}}{\Xi_t} = e^{-r_t^*}, \qquad r_{t+1}^* = \rho_r r_t^* + \sigma_r \varepsilon_{t+1}^r, \qquad \varepsilon_{t+1}^r \overset{\text{iid}}{\sim} \mathcal{N}(0,1),$$

where ε_{t+1}^r is independent of the policy innovation. The budget constraint is unchanged. The preference weight cancels from labor supply, while the bond first-order condition and the real discount factor become

$$C_t^{-\sigma} = \beta R_t e^{-r_t^*} \mathbb{E}_t \left[\frac{C_{t+1}^{-\sigma}}{\Pi_{t+1}} \right], \qquad M_{t,t+1} = \beta e^{-r_t^*} \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma}.$$

The same discount factor enters firms' price-setting condition (10). The remaining equilibrium conditions retain their form. Only the growth of Ξ_t affects decisions, so its level adds no state variable.

Second, replace equation (14) with

$$s_t^C = \varepsilon_{t+1}^{\theta} + \nu_R(\zeta_t)\eta_t^C, \qquad s_t = \zeta_t \varepsilon_{t+1}^{\theta} + \lambda_r(1-\zeta_t)\varepsilon_{t+1}^{r} + \nu_R(\zeta_t)\eta_t.$$

The Chair continues to communicate policy news. The speech's policy-news loading increases with similarity, while its demand-news loading decreases. For $0 \le \lambda_r \le 1$, let

$$v_s(\zeta) = \zeta^2 + \lambda_r^2(1-\zeta)^2, \qquad \nu_R^2(\zeta) = \frac{\sqrt{[v_s(\zeta)-1]^2+4} - [1+v_s(\zeta)]}{2}.$$

This nonnegative common noise variance preserves $\mathrm{Corr}(s_t^C, s_t \mid \zeta_t) = \zeta_t$: the covariance is ζ_t and the product of the signal variances is one. The same empirical similarity distribution therefore applies to both models.

Agents observe r_t^* along with the baseline observables and infer the two future innovations jointly. With

$$H(\zeta) = \begin{pmatrix} 1 & 0 \\ \zeta & \lambda_r(1-\zeta) \end{pmatrix}, \qquad S(\zeta) = H(\zeta)H(\zeta)' + \nu_R^2(\zeta) I_2,$$

the posterior mean and covariance are

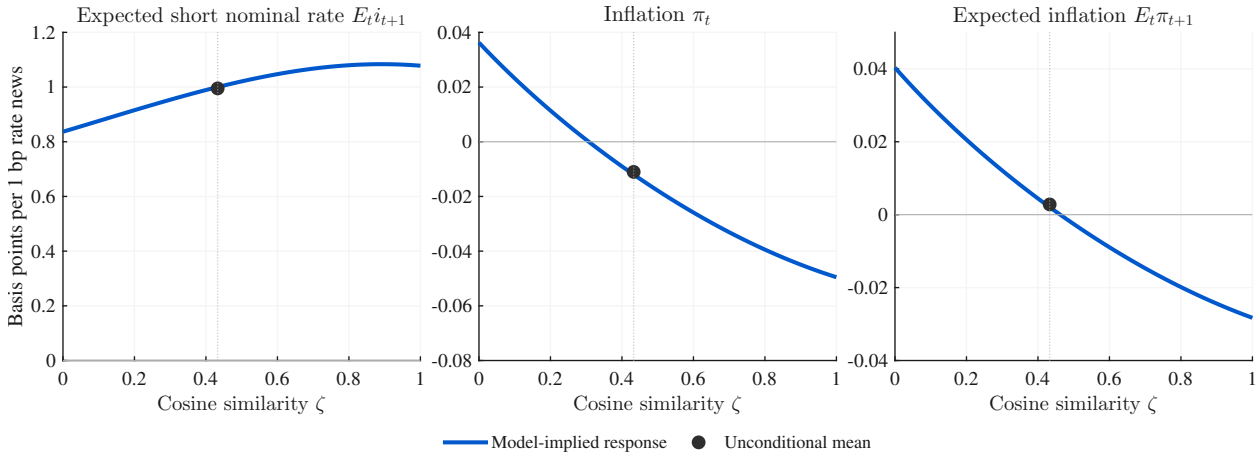
$$\boldsymbol{\mu}_t = K(\zeta_t) \begin{pmatrix} s_t^C \\ s_t \end{pmatrix}, \qquad K(\zeta) = H(\zeta)' S(\zeta)^{-1}, \qquad V_R(\zeta) = I_2 - K(\zeta)H(\zeta). \tag{17}$$

At $\zeta = 1$, the inverse is interpreted as the Moore–Penrose pseudoinverse. Consistent messages reveal the policy innovation, while the demand innovation remains unobserved, so that $V_R(1) = \mathrm{diag}(0,1)$. The recursive state becomes $X_t = (\theta_t, r_t^*, \zeta_t, \mu_t^{\theta}, \mu_t^{r})$. The equilibrium definition applies with this enlarged state and its transition law.

We set $\rho_r = 0.80$ and $\sigma_r = 0.0040$, allowing for persistent demand news, and calibrate $\lambda_r = 0.07$ to target a zero crossing of the model's expected-inflation response around 0.46–0.49, near the empirical crossing. The common macroeconomic parameters and the similarity distribution remain those of Table 6.

Figure 6 repeats the baseline experiment using the same positive messages. At low similarity, the member's speech raises expected demand sufficiently to offset the disinflationary policy news. At high similarity, the policy interpretation dominates and expected inflation falls. The nonlinear expected-inflation profile crosses zero near $\zeta = 0.46$. Thus, a rate in-

Figure 6: Similarity and the Composition of Communication News



Notes: The extension repeats the unit-message experiment with $\theta_t = r_t^* = 0$. Agents infer both future policy and demand innovations. The comparison sets both messages to zero at the same similarity. The blue response curves and black averages follow the conventions in Figure 5. The normalization uses the extension's own expected short-rate response at mean similarity.

crease expected after communication can coexist with either sign of the inflation response, depending on the news that agents infer. Appendix C reports the corresponding dynamic paths and their timing.

6 Conclusions

We study how coordination in FOMC communication affects monetary policy transmission. Using time-stamped speeches, we construct high-frequency monetary policy surprises and measure the similarity of each speech to the Chair's preceding post-meeting press conference. This allows us to ask whether subsequent speeches that reinforce the Chair's message are redundant for financial markets or provide additional information that strengthens monetary policy transmission.

Our results favor the latter interpretation. On average, speech-induced monetary policy surprises have little effect on inflation expectations or equity prices. When communication is more coordinated, however, a tightening surprise raises short-term interest rates while lowering inflation expectations and equity prices. Aligned speeches therefore continue to convey information to markets even when they closely echo the Chair's preceding message. At lower levels of similarity, these responses weaken and can reverse, consistent with speeches conveying additional information beyond the common policy message.

An imperfect-information New Keynesian model rationalizes these findings. When the Chair and an FOMC participant provide aligned signals about future policy, agents infer the common policy message more precisely and monetary policy transmission becomes stronger.

Allowing less aligned speeches to convey information about future economic fundamentals can account for the different market responses observed at low similarity.

The heterogeneity documented across policymakers is a natural feature of committee-based monetary policy. Our findings show that when policymakers' messages become more coordinated, they need not become redundant. Subsequent aligned communication can instead convey incremental information that strengthens the transmission of monetary policy.

References

Acosta, M., A. Ajello, M. D. Bauer, F. Loria, and S. Miranda-Agrippino (2026). Financial market effects of FOMC communication: Evidence from a new event-study database. Working Paper 2025-30, Federal Reserve Bank of San Francisco. Revised August 2026; revise and resubmit at the Journal of Monetary Economics.

Ahrens, M. and M. McMahon (2021). Extracting economic signals from central bank speeches. In *Proceedings of the Third Workshop on Economics and Natural Language Processing*, pp. 93–114. Association for Computational Linguistics.

Altavilla, C., R. S. Gürkaynak, T. Kind, and L. Laeven (2026). Monetary transmission with frequent policy events. *Journal of Monetary Economics 163*, 103986.

Andrade, P. and F. Ferroni (2021). Delphic and odyssean monetary policy shocks: Evidence from the euro area. *Journal of Monetary Economics 117*, 816–832.

Aruoba, S. B. and T. Drechsel (2026, forthcoming). Identifying monetary policy shocks: A natural language approach. *American Economic Journal: Macroeconomics*.

Bauer, M. D. and E. T. Swanson (2023a, March). An alternative explanation for the "fed information effect". *American Economic Review 113*(3), 664–700.

Bauer, M. D. and E. T. Swanson (2023b). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual 37*, 87–155.

Blinder, A. S., M. Ehrmann, M. Fratzscher, J. De Haan, and D.-J. Jansen (2008). Central bank communication and monetary policy: A survey of theory and evidence. *Journal of Economic Literature 46*(4), 910–945.

Bordo, M. D. and K. Istrefi (2023). Perceived FOMC: The making of hawks, doves and swingers. *Journal of Monetary Economics*.

Campbell, J. R., C. L. Evans, J. D. M. Fisher, and A. Justiniano (2012). Macroeconomic effects of federal reserve forward guidance. *Brookings Papers on Economic Activity 2012*(1), 1–80.

Campbell, J. R., J. D. M. Fisher, A. Justiniano, and L. Melosi (2017). Forward guidance and macroeconomic outcomes since the financial crisis. *NBER Macroeconomics Annual 31*,

283–357.

Cieslak, A. and A. Schrimpf (2019). Non-monetary news in central bank communication. *Journal of International Economics 118*, 293–315.

Cochrane, J. H. and M. Piazzesi (2002). The fed and interest rates—a high-frequency identification. *American Economic Review 92*(2), 90–95.

Ehrmann, M. and M. Fratzscher (2007). Communication by central bank committee members: Different strategies, same effectiveness? *Journal of Money, Credit and Banking 39*(2-3), 509–541.

Ehrmann, M. and M. Fratzscher (2013). Dispersed communication by central bank committees and the predictability of monetary policy decisions. *Public Choice 157*(1–2), 223–244.

Ehrmann, M., R. Tietz, and B. Visser (2021, June). Voting right rotation, behavior of committee members and financial market reactions: evidence from the U.S. Federal Open Market Committee. Working Paper Series 2569, European Central Bank.

Gorodnichenko, Y., T. Pham, and O. Talavera (2023). The voice of monetary policy. *American Economic Review 113*(2), 548–584.

Granziera, E., V. H. Larsen, G. Meggiorini, and L. Melosi (2025). Speaking of inflation: The influence of fed speeches on expectations. Discussion Paper 20038, CEPR.

Gürkaynak, R. S., B. P. Sack, and E. T. Swanson (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking 1*(1), 55–93.

Handlan, A. (2022a, September). Fedspeak matters: Statement similarity and monetary policy expectations. Working paper, Brown University.

Handlan, A. (2022b). Text shocks and monetary surprises: Text analysis of fomc statements with machine learning. Working paper.

Hansen, S. and M. McMahon (2016a). First impressions matter: Signalling as a source of policy dynamics. *The Review of Economic Studies 83*(4), 1645–1672.

Hansen, S. and M. McMahon (2016b). Shocking language: Understanding the macroeconomic effects of central bank communication. *Journal of International Economics 99*(S1), S114–S133.

Hansen, S., M. McMahon, and A. Prat (2018). Transparency and deliberation within the FOMC: A computational linguistics approach. *Quarterly Journal of Economics 133*(2), 801–870.

Istrefi, K., F. Odendahl, and G. Sestieri (2023). Fed communication on financial stability concerns and monetary policy decisions: Revelations from speeches. *Journal of Banking Finance 151*, 106820.

Jarociński, M. and P. Karadi (2020). Deconstructing monetary policy surprises—the role of

information shocks. *American Economic Journal: Macroeconomics 12*(2), 1–43.

Kuttner, K. N. (2001). Monetary policy surprises and interest rates: Evidence from the fed funds futures market. *Journal of Monetary Economics 47*(3), 523–544.

Malmendier, U., S. Nagel, and Z. Yan (2021). The making of hawks and doves. *Journal of Monetary Economics 117*, 19–42.

Melosi, L. (2017). Signalling effects of monetary policy. *The Review of Economic Studies 84*(2), 853–884.

Miranda-Agrippino, S. and G. Ricco (2021). The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics 13*(3), 74–107.

Nakamura, E. and J. Steinsson (2018). High-frequency identification of monetary non-neutrality. *Quarterly Journal of Economics 133*(3), 1283–1330.

Neuhierl, A. and M. Weber (2019). Monetary policy communication, policy slope, and the stock market. *Journal of Monetary Economics 108*, 140–155.

Swanson, E. T. (2023, May). The importance of fed chair speeches as a monetary policy tool. *AEA Papers and Proceedings 113*, 394–400.

Swanson, E. T. and V. Jayawickrema (2024). Speeches by the fed chair are more important than fomc announcements: An improved high-frequency measure of us monetary policy shocks. Working Paper.

Vissing-Jorgensen, A. (2019). Central banking with many voices: The communications arms race. In *XXIII Annual Conference of the Central Bank of Chile*.

Woodford, M. (2005). Central bank communication and policy effectiveness. *NBER Working Paper* (11898).

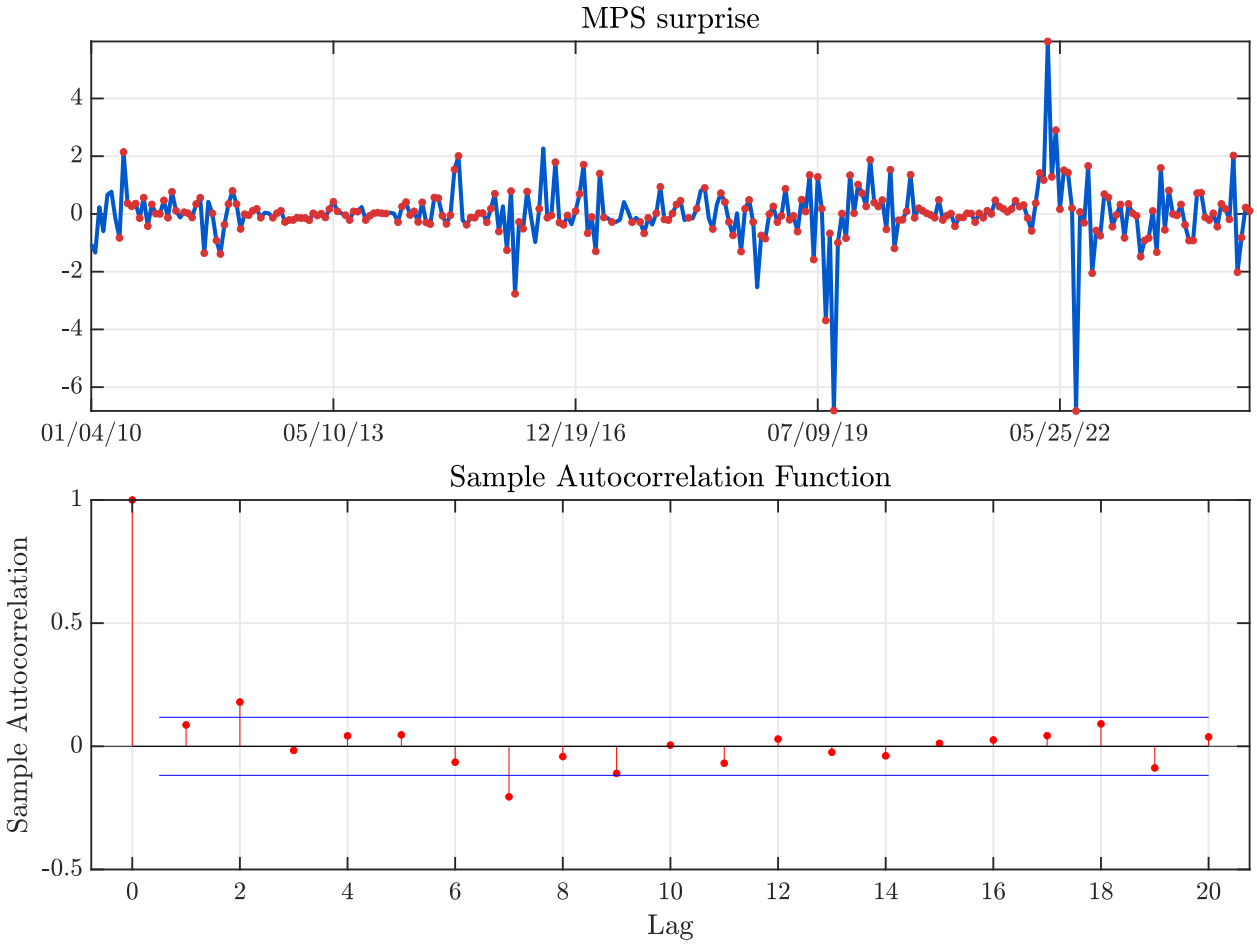
A Appendix

Table A1: Individual speaker fixed effects

Speaker	Estimate	SE OLS	SE two-way cluster	t two-way cluster	p two-way cluster
Bernanke	0.1761	0.0470	0.0395	4.464	0.0003
Bowman	0.1020	0.0315	0.0028	36.988	< 0.0001
Brainard	0.1373	0.0364	0.0251	5.462	< 0.0001
Clarida	0.2197	0.0408	0.0167	13.128	< 0.0001
Cook	0.1519	0.0381	0.0228	6.671	< 0.0001
Duke	0.1039	0.0553	0.0367	2.831	0.0107
Fischer	0.2128	0.0434	0.0369	5.765	< 0.0001
Jefferson	0.1890	0.0441	–	–	–
Kashkari	0.2626	0.0433	0.0361	7.282	< 0.0001
Kugler	0.2529	0.0523	0.0082	30.776	< 0.0001
Powell	0.1701	0.0354	0.0222	7.674	< 0.0001
Quarles	0.0931	0.0395	0.0199	4.681	0.0002
Raskin	0.1469	0.0575	0.0419	3.506	0.0024
Rosengren	0.2403	0.0410	0.0296	8.129	< 0.0001
Stein	0.1812	0.0541	0.0249	7.289	< 0.0001
Tarullo	0.0781	0.0453	0.0294	2.656	0.0156
Waller	0.2022	0.0343	0.0210	9.636	< 0.0001
Williams	0.3622	0.0752	0.0332	10.905	< 0.0001
Yellen	0.2131	0.0422	0.0308	6.926	< 0.0001

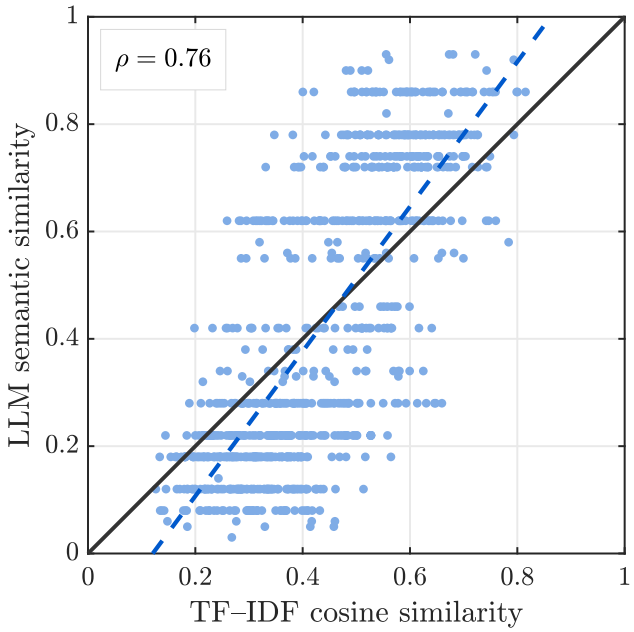
Notes: The omitted speaker is Barr. Estimates are relative to this reference speaker and come from the full specification.

Figure A1: Monetary Policy Speech (MPS) Surprise.



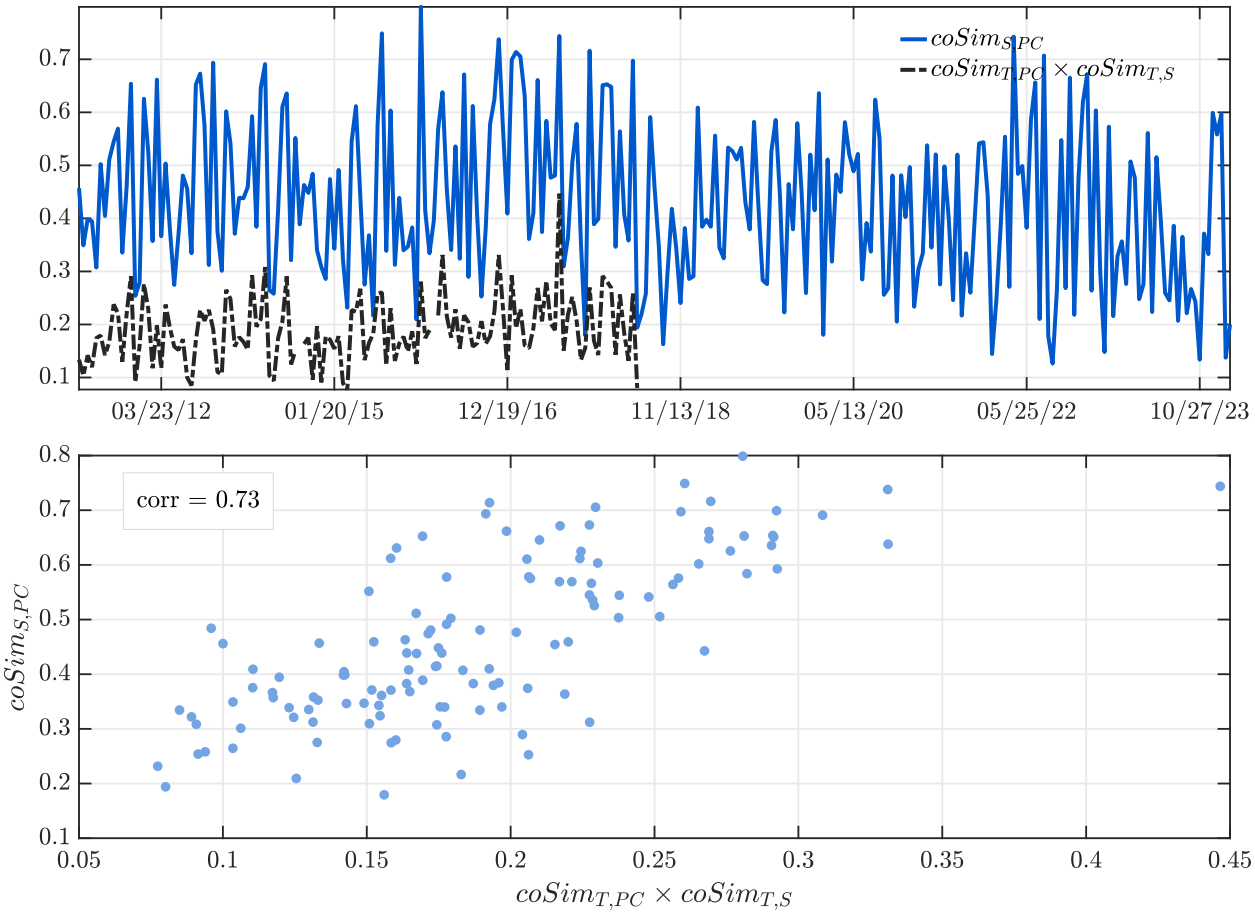
Notes: The top panel plots the monetary policy speech (MPS) surprise constructed from five-minute changes in eurodollar futures around each speech. The high-frequency speech-surprise series is plotted from 2010 onward, while the similarity-based analysis starts in April 2011, when FOMC press conferences became available. The blue line shows the MPS series for all speeches; red markers indicate speeches delivered by Board of Governors (BoG) members. The bottom panel reports the sample autocorrelation function of the MPS series; horizontal lines denote 95% confidence bands under the null of zero autocorrelation.

Figure A2: Similarity between individual speeches and press conferences: Baseline measure vs. an LLM-derived one.



Notes: This figure plots TF–IDF cosine similarity between FOMC member speeches and the Chair's post-meeting press conference (x-axis) against an LLM-based semantic similarity score (y-axis). Each point corresponds to a speech–press-conference pair. The solid black line is the 45-degree line. The in-panel value reports the sample correlation between the two measures ($\rho = 0.76$). The horizontal banding reflects discretization of the LLM scores (e.g., rounding or use of focal values in the model's numeric outputs).

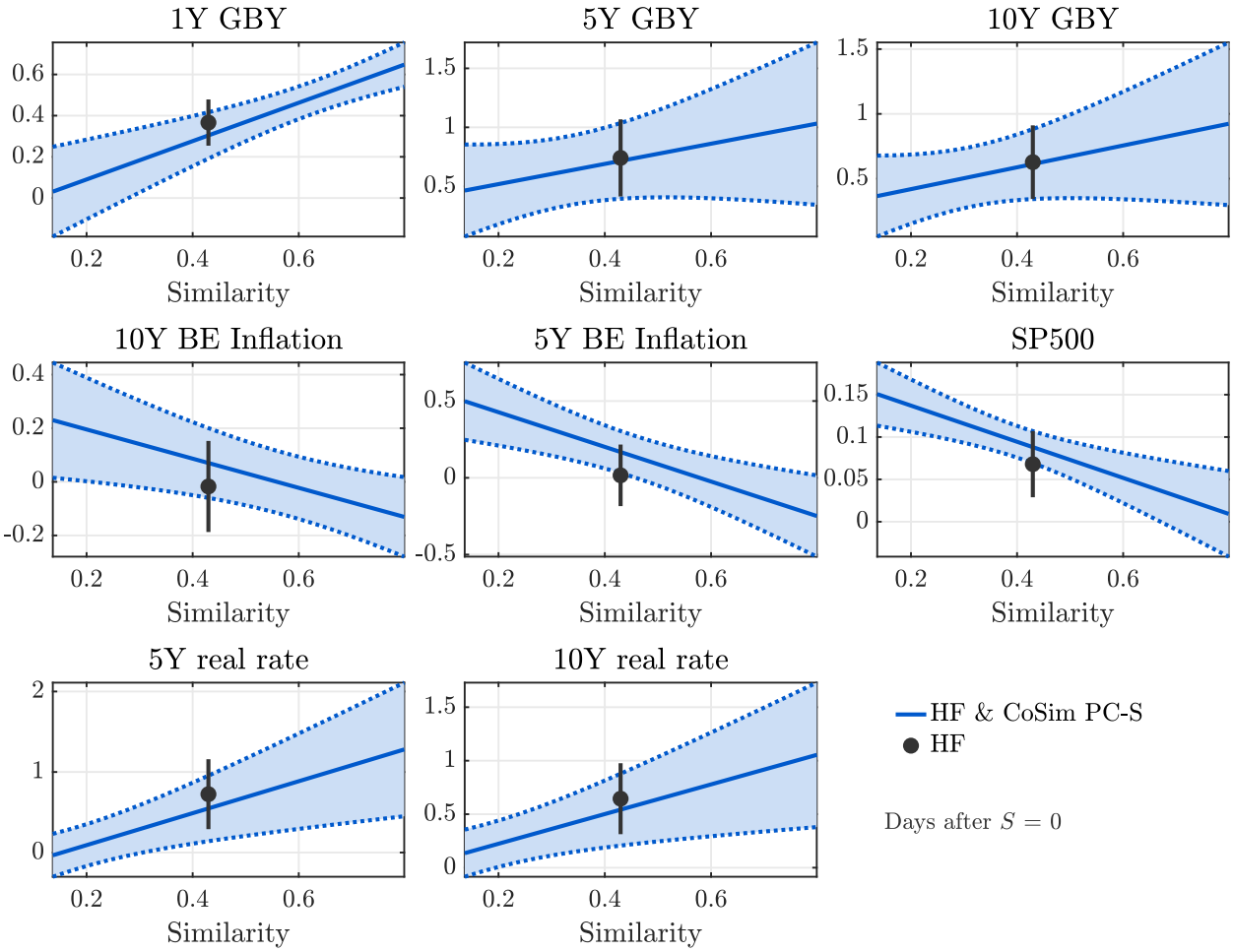
Figure A3: Similarity between individual speeches and press conferences: Baseline measure vs. filtering out non-FOMC-relevant topics.



Notes: The top panel plots two measures of textual alignment with the Chair's post-meeting press conference. The solid blue line is the baseline cosine similarity between each speech and the preceding Chair press conference, $coSim_{S,PC}$. The dash–dot black line is an alternative measure that down-weights speeches less related to contemporaneous FOMC deliberations, defined as $coSim_{T,PC} \times coSim_{T,S}$, where T denotes the transcript of the corresponding FOMC meeting; because transcripts are available only through 2018, this series ends in 2018. The bottom panel plots $coSim_{S,PC}$ against $coSim_{T,PC} \times coSim_{T,S}$ for speeches with available transcripts; the in-panel value reports their sample correlation.

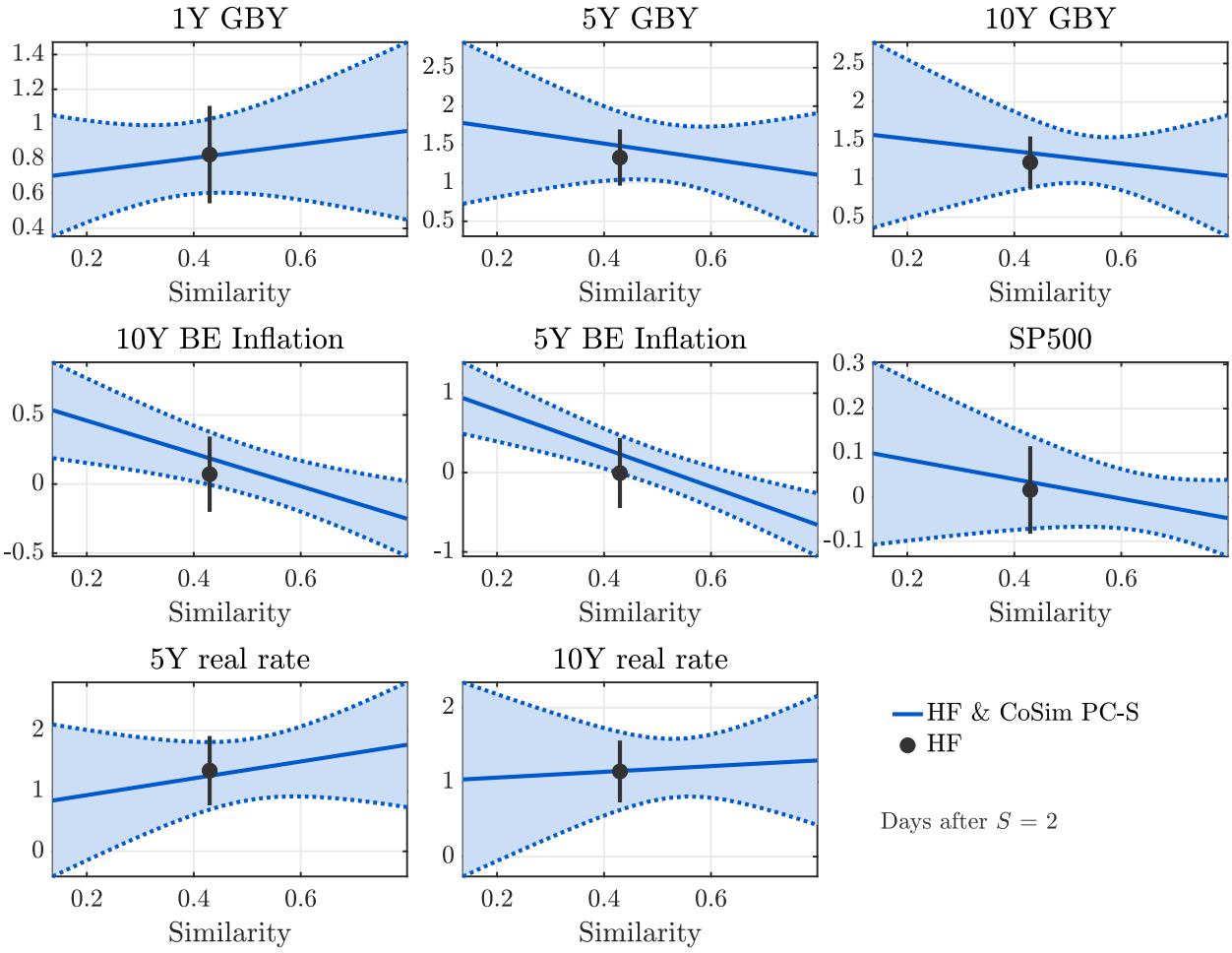
A.1 Robustness Results: Changing the Time Horizon

Figure A4: Financial-Market Responses to MPS Surprises across Speech–Press-Conference Similarity ($h = 0$).



Notes: Each panel reports the estimated marginal effect of a tightening monetary policy speech (MPS) surprise on the indicated financial variable, as a function of the speech–press-conference similarity index $\ell_t \in [0, 1]$, measured at horizon $h = 0$. The solid blue line plots $\beta_h + \delta_n \ell_t$ from equation (4). Shaded areas and dotted lines denote pointwise confidence bands. The black circle indicates the unconditional effect β_h estimated without the interaction term, evaluated at the sample mean of ℓ_t; vertical bars report its confidence interval.

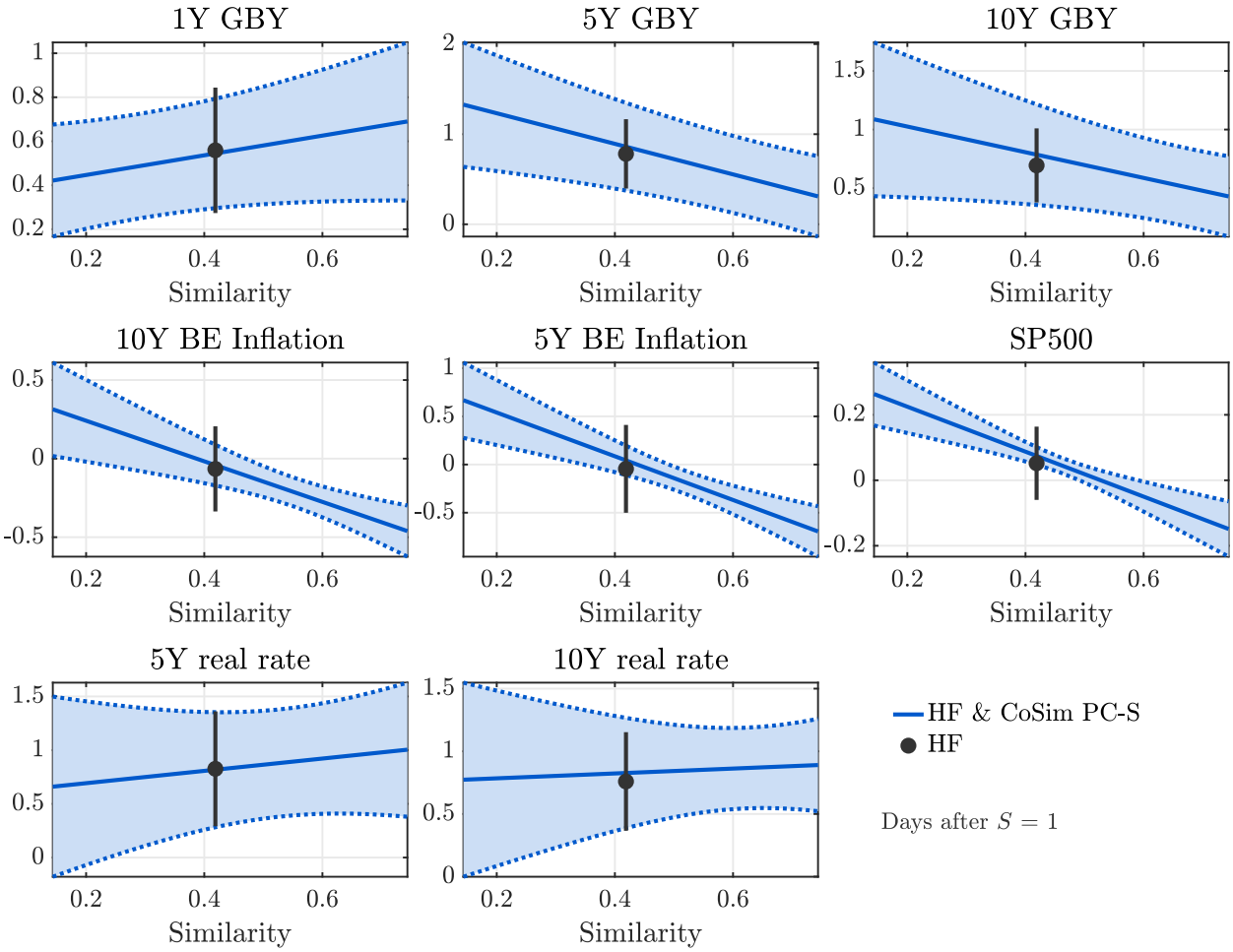
Figure A5: Financial-Market Responses to MPS Surprises across Speech–Press-Conference Similarity ($h = 2$).



Notes: Each panel reports the estimated marginal effect of a tightening monetary policy speech (MPS) surprise on the indicated financial variable, as a function of the speech–press-conference similarity index $\ell_t \in [0, 1]$, measured at horizon $h = 2$. The solid blue line plots $\beta_h + \delta_n \ell_t$ from equation (4). Shaded areas and dotted lines denote pointwise confidence bands. The black circle indicates the unconditional effect β_h estimated without the interaction term, evaluated at the sample mean of ℓ_t; vertical bars report its confidence interval.

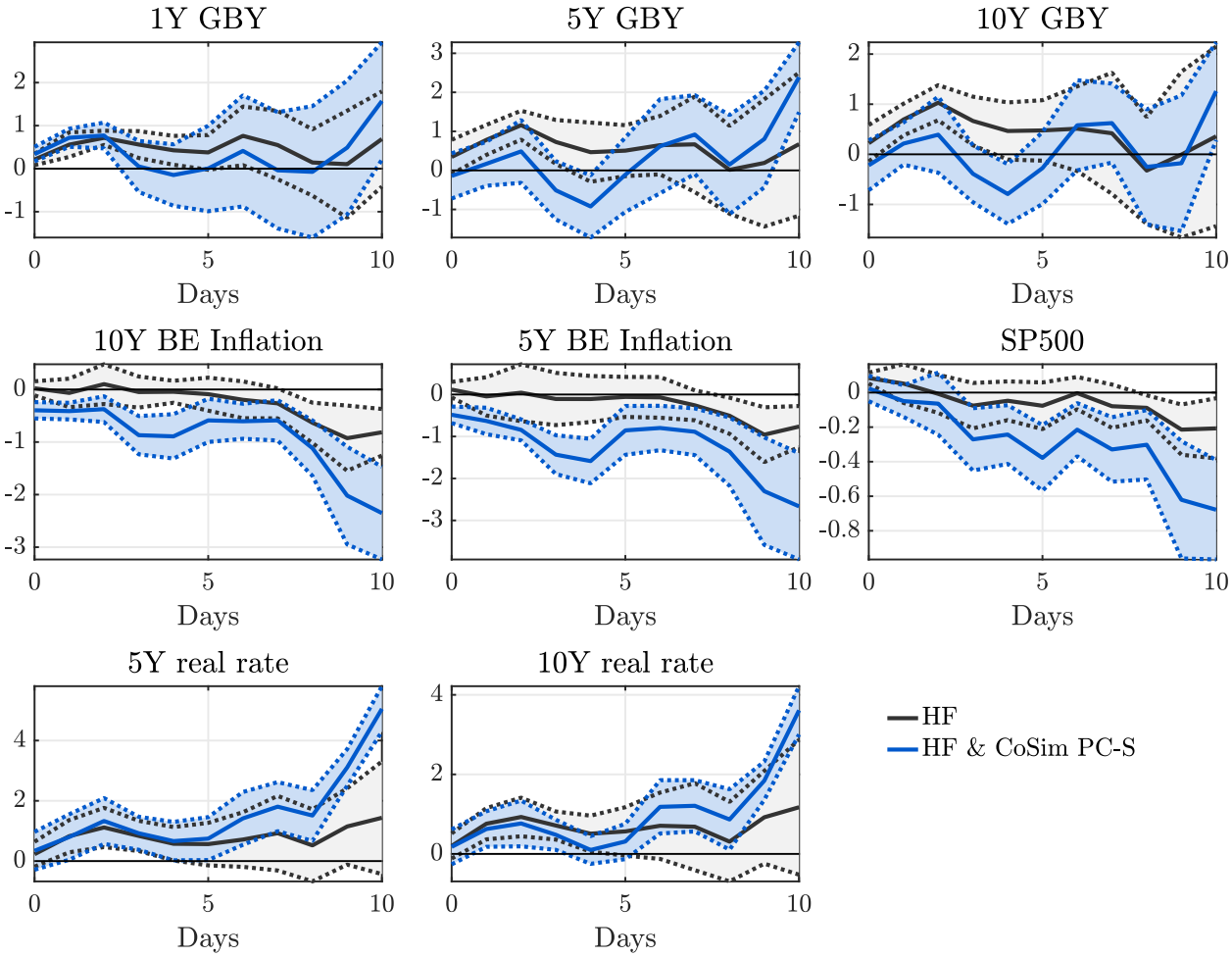
A.2 Robustness Results: Excluding Chair Speeches

Figure A6: Responses to MPS Surprises by Speech–Press-Conference Similarity (Excluding Chair Speeches).



Notes: This figure shows the response of financial variables to a tightening monetary policy speech (MPS) surprise across different levels of textual similarity between the speech and the Chair's most recent post-meeting press conference, measured one day after the speech. The solid blue line reports the estimated marginal effect as a function of similarity; dotted blue lines denote pointwise confidence bands. Black circles indicate the unconditional (average) effect estimated without interacting the MPS surprise with the similarity measure. The sample excludes speeches delivered by the FOMC Chair.

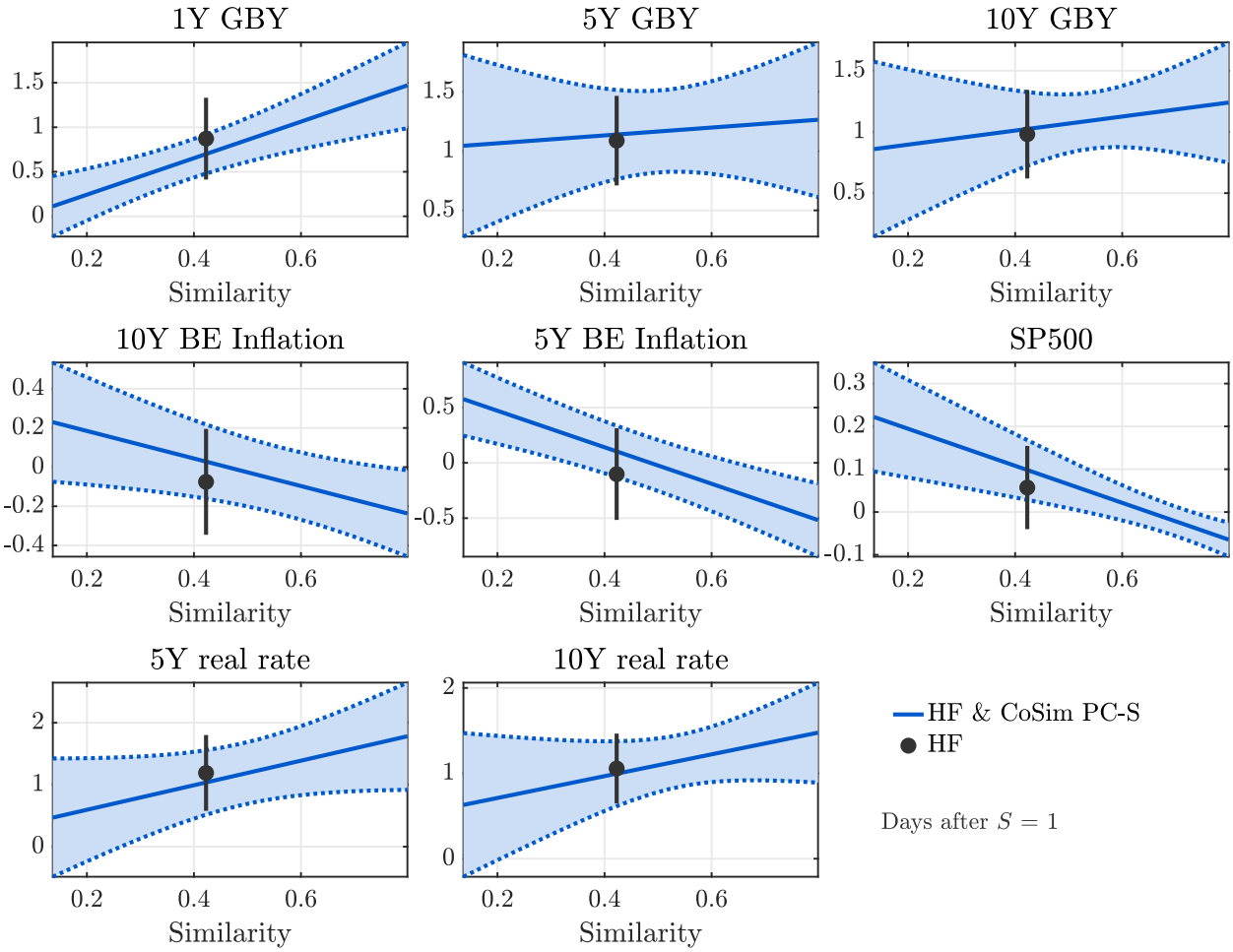
Figure A7: Dynamic Responses to MPS Surprises (High Similarity Speeches, Excluding Chair Speeches).



Notes: This figure reports daily local-projection impulse responses of financial variables to a tightening monetary policy speech (MPS) surprise. The blue line shows responses estimated using only speeches whose speech–press-conference cosine similarity exceeds one standard deviation above its sample mean; the black line shows the baseline response using all speeches. Dotted lines denote pointwise confidence bands around each response. The sample excludes speeches delivered by the FOMC Chair.

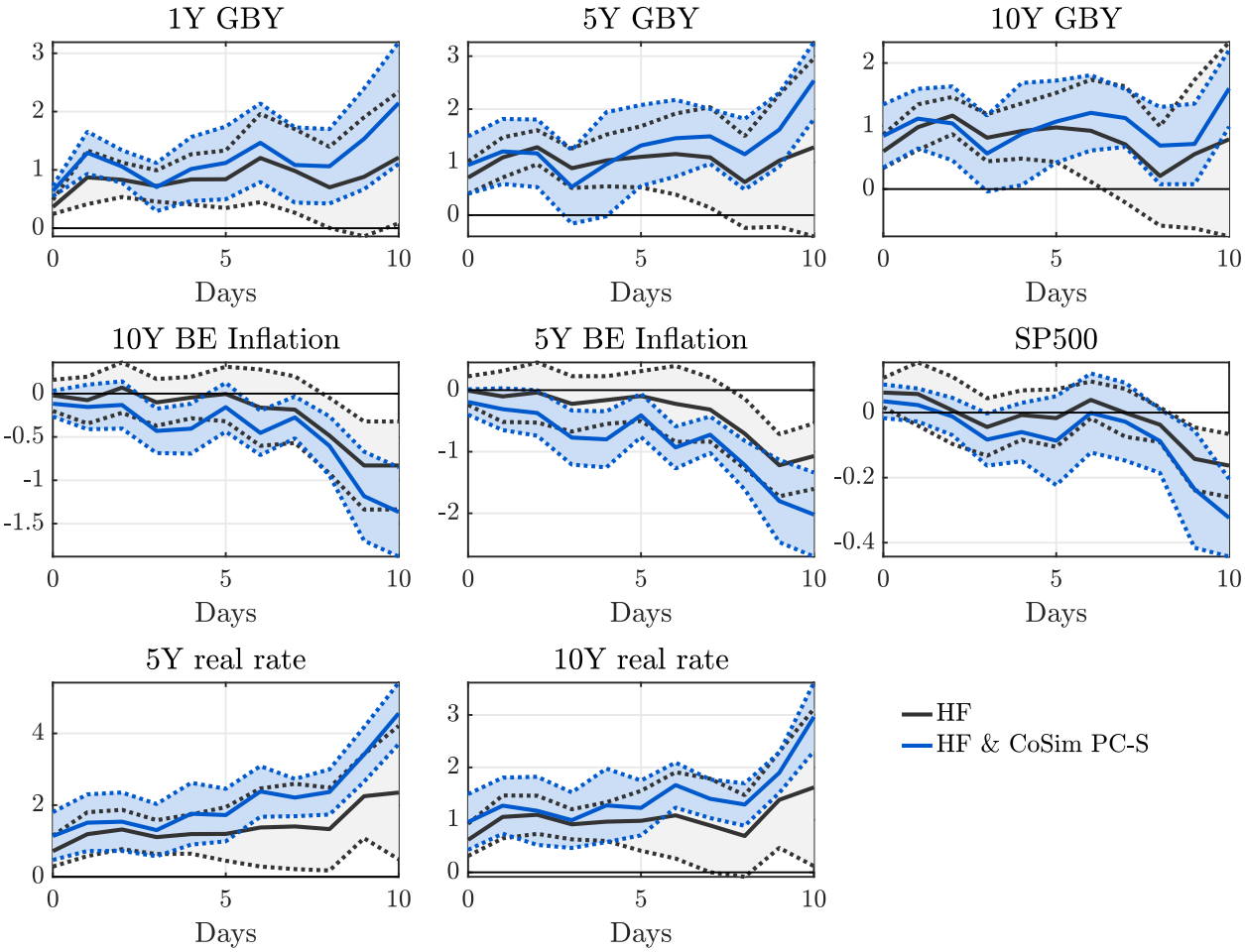
A.3 Robustness Results: Excluding Nonvoter Speeches

Figure A8: Responses to MPS Surprises by Speech–Press-Conference Similarity (Excluding Nonvoter Speeches).



Notes: This figure shows the response of financial variables to a tightening monetary policy speech (MPS) surprise across different levels of textual similarity between the speech and the Chair's most recent post-meeting press conference, measured one day after the speech. The solid blue line reports the estimated marginal effect as a function of similarity; dotted blue lines denote pointwise confidence bands. Black circles indicate the unconditional (average) effect estimated without interacting the MPS surprise with the similarity measure. The sample excludes speeches delivered by FOMC members without voting rights.

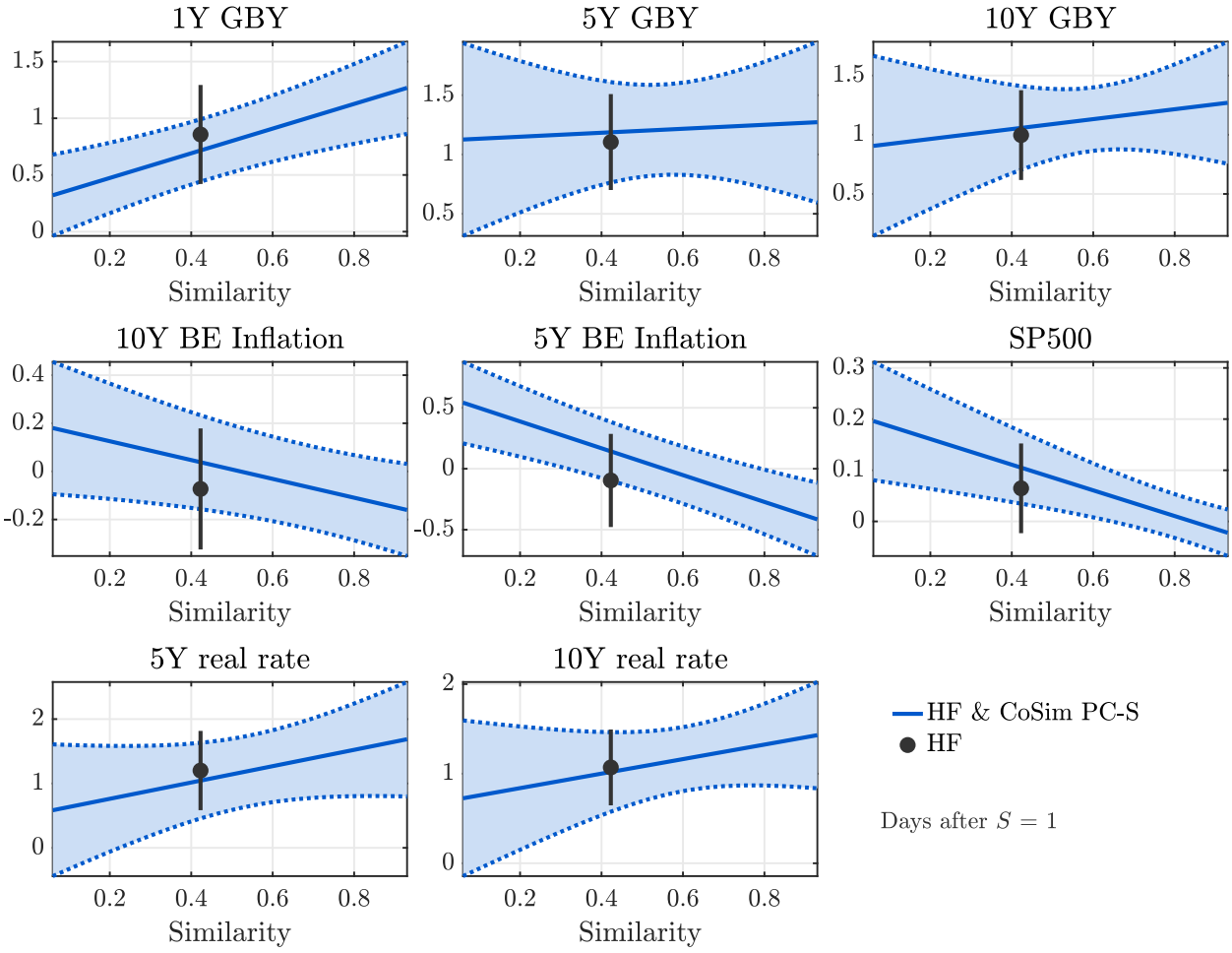
Figure A9: Dynamic Responses to MPS Surprises (High Similarity Speeches, Excluding Nonvoter Speeches).



Notes: This figure reports daily local-projection impulse responses of financial variables to a tightening monetary policy speech (MPS) surprise. The blue line shows responses estimated using only speeches whose speech–press-conference cosine similarity exceeds one standard deviation above its sample mean; the black line shows the baseline response using all speeches. Dotted lines denote pointwise confidence bands around each response. The sample excludes speeches delivered by FOMC members without voting rights.

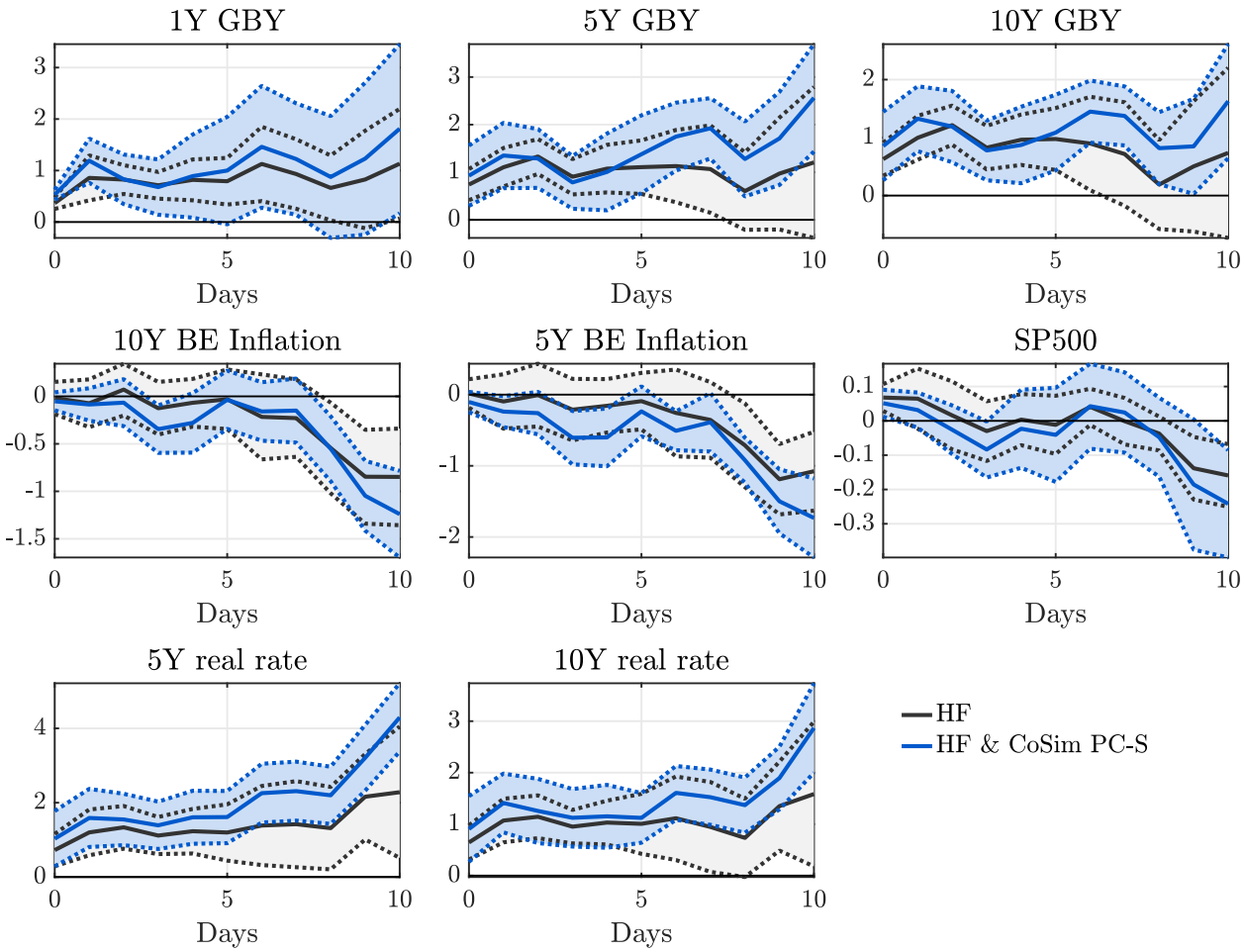
A.4 Robustness Results: LLM-Based Similarity Measure

Figure A10: LLM-Based Similarity and Financial-Market Responses to Monetary Policy Speech Surprises ($h = 1$).



Notes: Each panel plots the estimated effect of a tightening monetary policy speech (MPS) surprise on the indicated financial variable as a function of LLM-based semantic similarity between the speech and the Chair's most recent post-meeting press conference (horizontal axis). The solid blue line shows the fitted marginal effect $\beta_h + \delta_h \tilde{\ell}_t$ evaluated at horizon $h = 1$ (one day after the speech); shaded areas (and dotted boundaries) report the corresponding confidence band. Black circles denote the unconditional effect estimated without interacting the MPS surprise with similarity. The legend distinguishes the unconditional (HF) effect from the similarity-conditioned (HF & LLM similarity) effect.

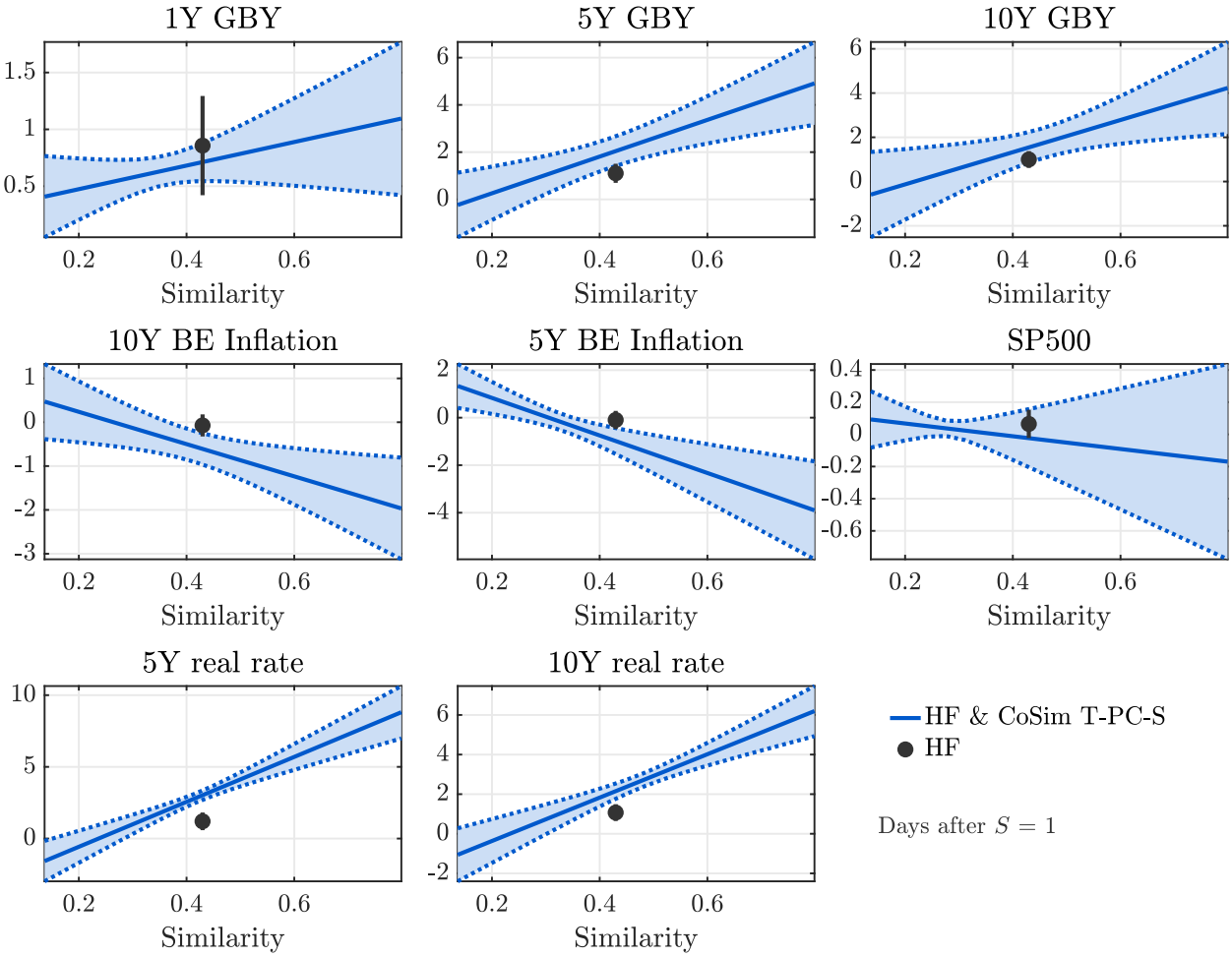
Figure A11: Dynamic Responses to Speech Surprises: High LLM-Based Similarity Sample.



Notes: This figure reports daily local-projection responses of financial variables to a tightening monetary policy speech (MPS) surprise over horizons $h = 0, \ldots, 10$. The blue line isolates responses using only speeches with high LLM-based semantic similarity to the Chair's most recent post-meeting press conference (similarity above one standard deviation from the sample mean); the gray/black line reports the corresponding response using all speeches. Shaded areas and dotted lines denote confidence bands around the respective responses. The bottom-right tile is reserved for the legend.

A.5 Robustness Results: Filtering Out Non-FOMC-Relevant Topics

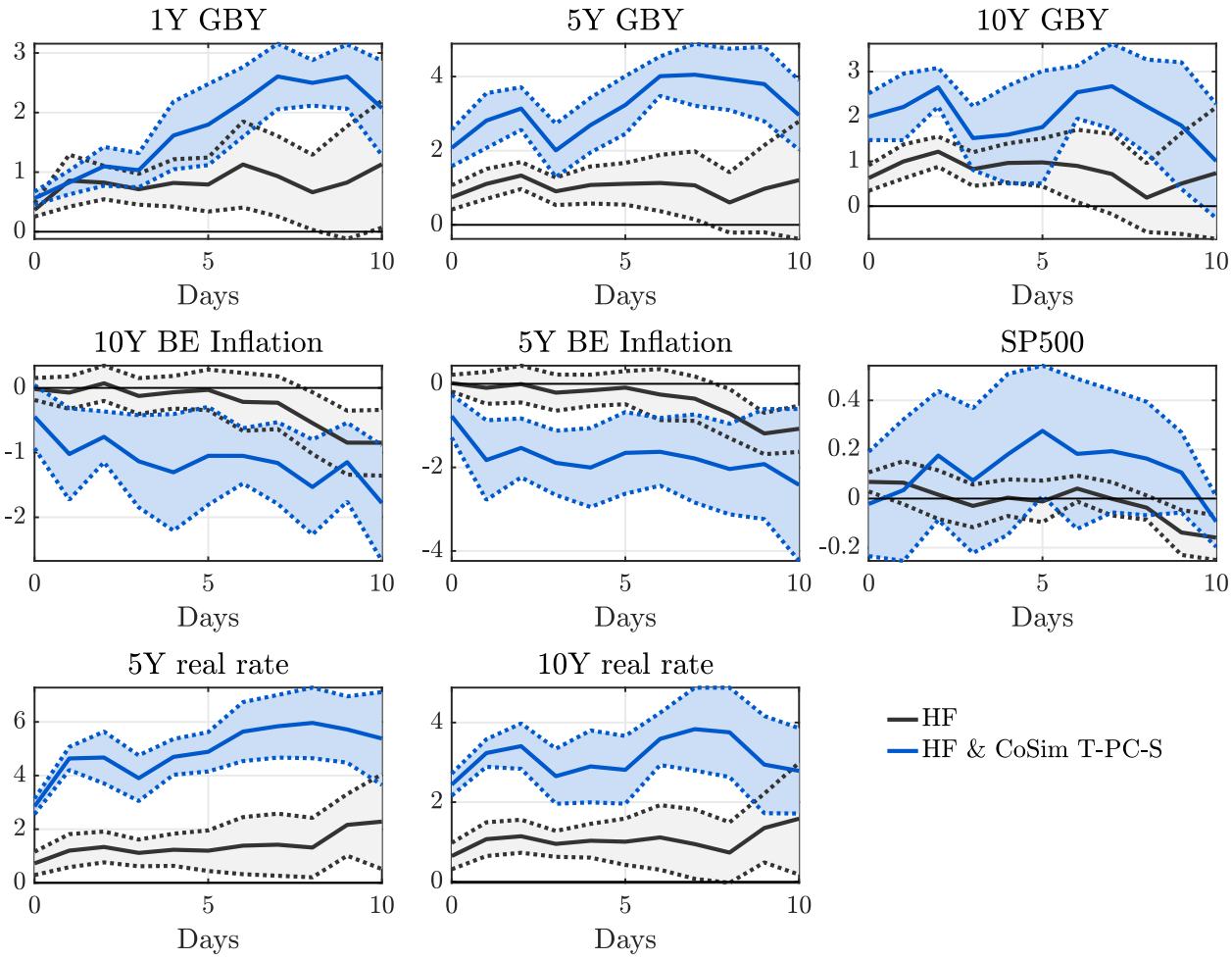
Figure A12: Financial-Market Responses by Similarity to the Chair and the FOMC Discussion ($h = 1$).



Notes: Each panel plots the estimated effect of a tightening monetary policy speech (MPS) surprise on the indicated financial variable as a function of the combined similarity between the speech, the transcript of the FOMC discussion, and the Chair's most recent post-meeting press conference (horizontal axis). The solid blue line shows the fitted marginal effect $\beta_h + \delta_h \tilde{\ell}_t$ evaluated at horizon $h = 1$ (one day after the speech); shaded areas (and dotted boundaries) report the corresponding confidence band. Black circles denote the unconditional effect estimated without interacting the MPS surprise with similarity. The legend distinguishes the unconditional (HF) effect from the similarity-conditioned (HF & combined similarity) effect.

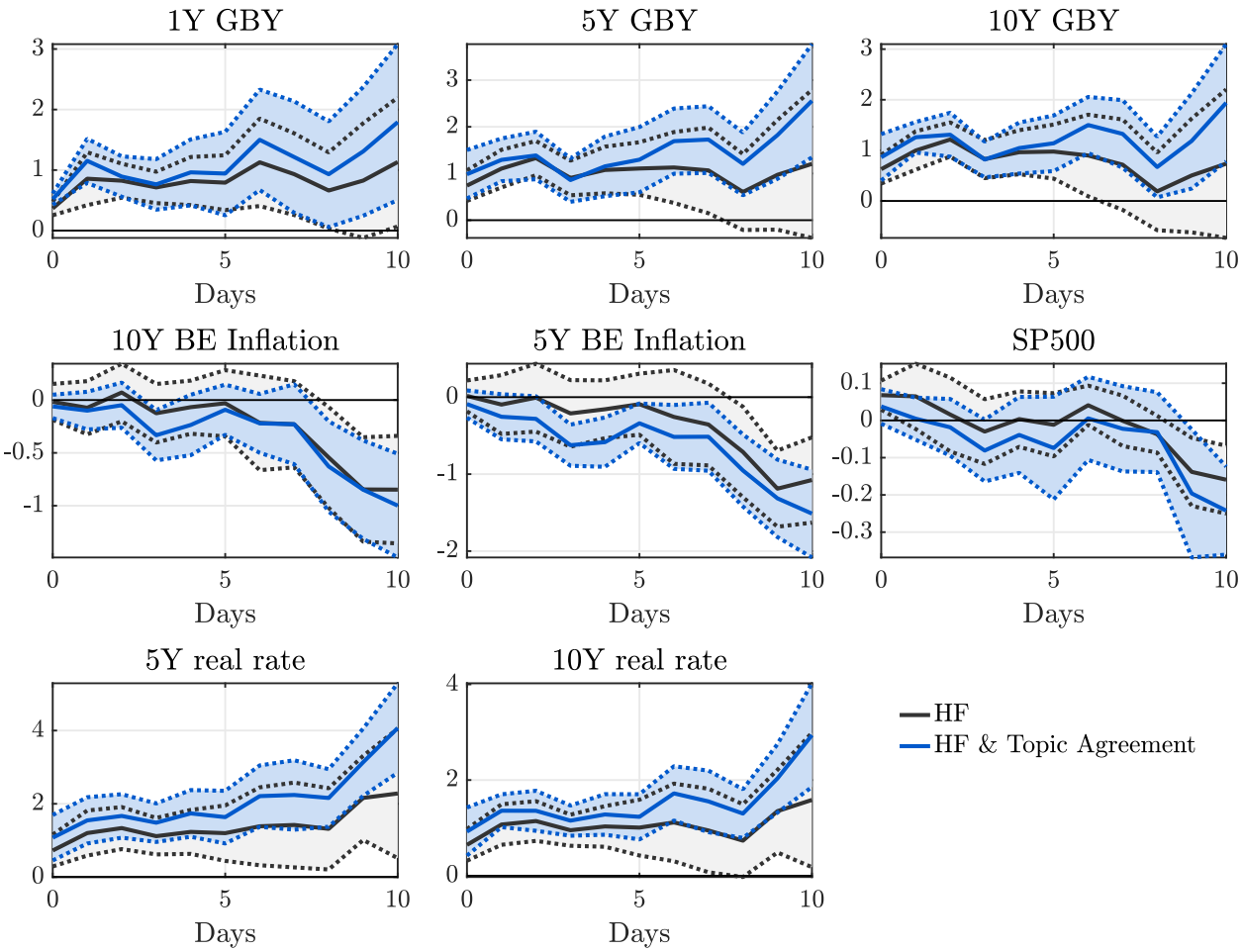
A.6 Robustness Results: "Yes-Yes" subcells of Figure 2

Figure A13: Dynamic Responses to Speech Surprises: High similarity to the Chair and the FOMC Discussion.



Notes: This figure reports daily local-projection responses of financial variables to a tightening monetary policy speech (MPS) surprise over horizons $h = 0, \ldots, 10$. The blue line isolates responses using only speeches with high combined similarity between the speech, the transcript of the FOMC discussion and the Chair's most recent post-meeting press conference (similarity above one standard deviation from the sample mean); the gray/black line reports the corresponding response using all speeches. Shaded areas and dotted lines denote confidence bands around the respective responses. The bottom-right tile is reserved for the legend.

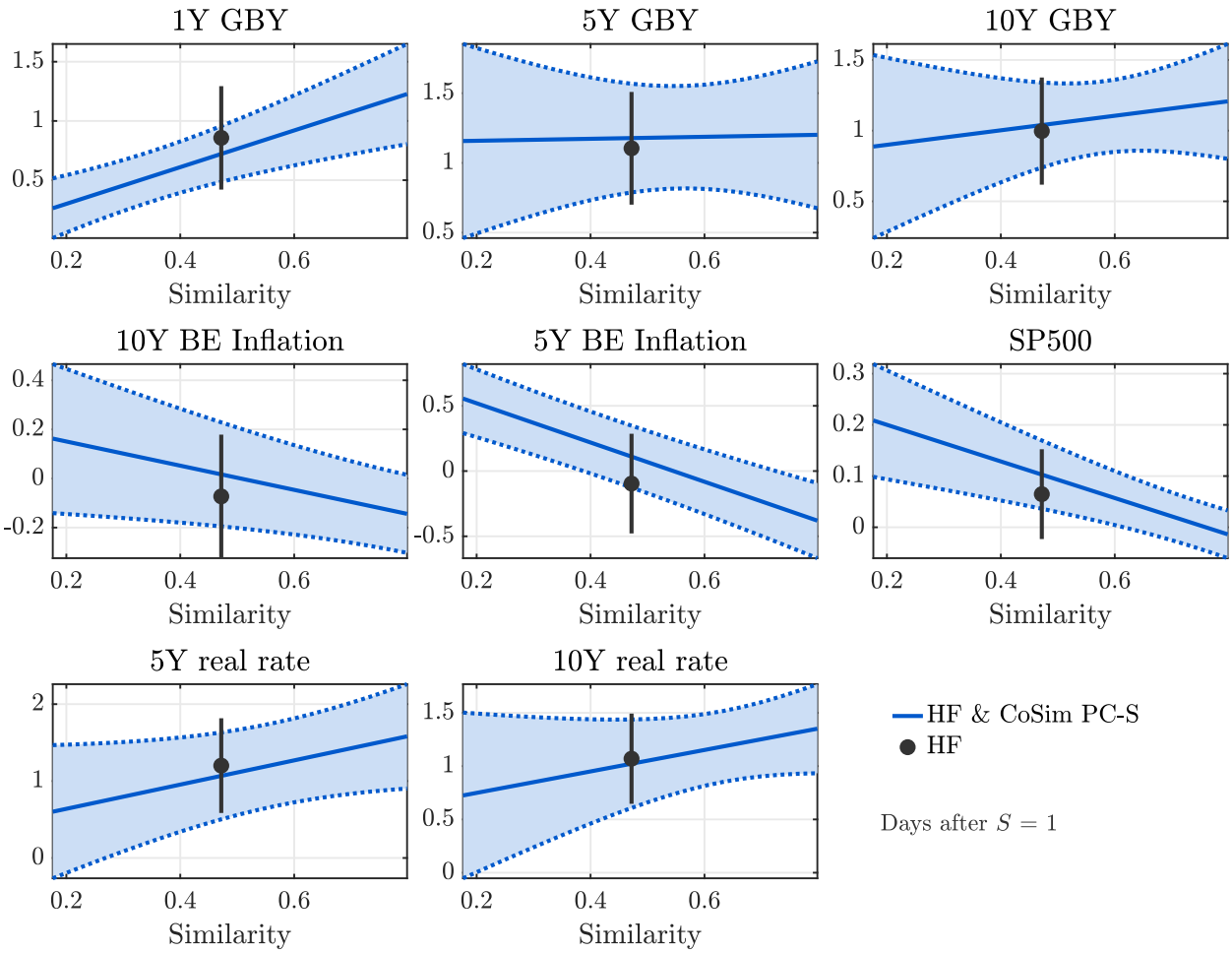
Figure A14: Dynamic Responses to Speech Surprises: "Yes-Yes" subcells of Figure 2.



Notes: This figure reports daily local-projection responses of financial variables to a tightening monetary policy speech (MPS) surprise over horizons $h = 0, \ldots, 10$. The blue line isolates responses using only speeches that discuss the same topic and in which the speakers agree with each other. The classification scheme follows the LLM procedure summarized by Figure 2. The gray/black line reports the corresponding response using all speeches. Shaded areas and dotted lines denote confidence bands around the respective responses. The bottom-right tile is reserved for the legend.

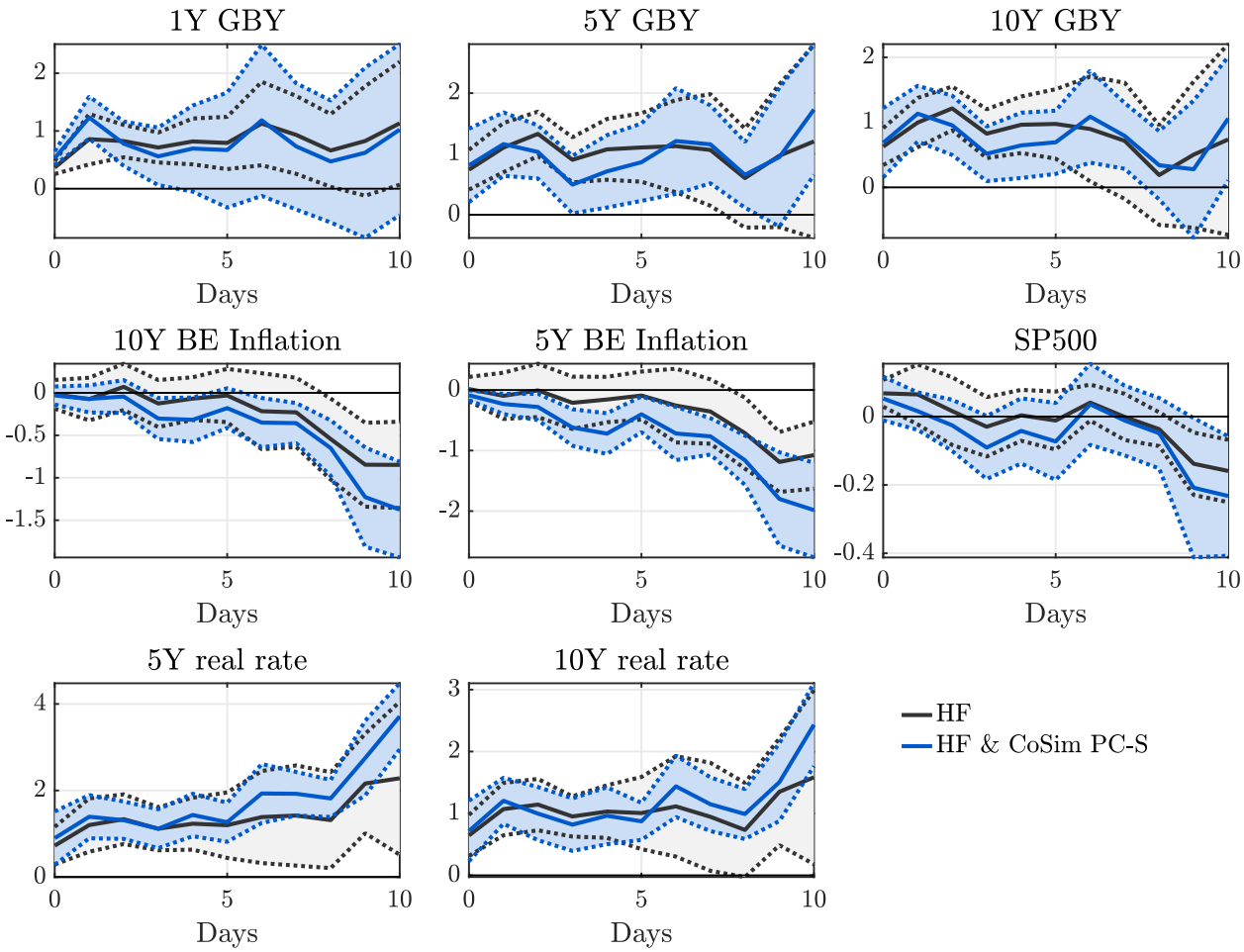
A.7 Robustness Results: Press Conference without Q&A

Figure A15: Press Conference without Q&A and Financial-Market Responses to Monetary Policy Speech Surprises ($h = 1$).



Notes: Each panel plots the estimated effect of a tightening monetary policy speech (MPS) surprise on the indicated financial variable as a function of semantic similarity between the speech and the Chair's most recent post-meeting press conference (horizontal axis) without Q&A. The solid blue line shows the fitted marginal effect $\beta_h + \delta_h \tilde{\ell}_t$ evaluated at $h = 1$ day after the speech; shaded areas (and dotted boundaries) report the corresponding confidence band. Black circles denote the unconditional effect estimated without interacting the MPS surprise with similarity. The legend distinguishes the unconditional (HF) effect from the similarity-conditioned effect.

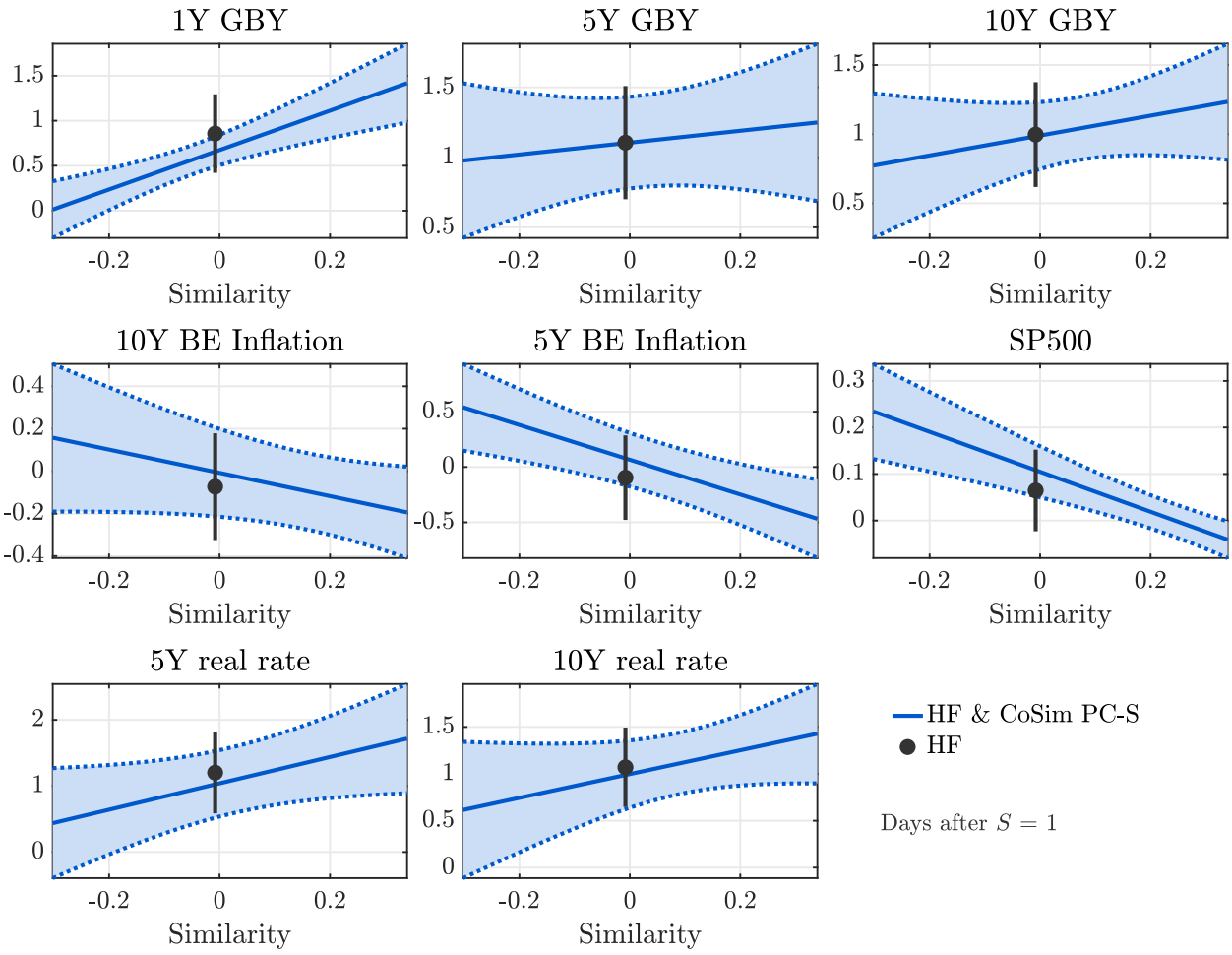
Figure A16: Dynamic Responses to Speech Surprises: Press Conference without Q&A.



Notes: This figure reports daily local-projection responses of financial variables to a tightening monetary policy speech (MPS) surprise over horizons $h = 0, \ldots, 10$. The blue line isolates responses using only speeches with high semantic similarity to the Chair's most recent post-meeting press conference without Q&A (similarity above one standard deviation from the sample mean); the gray/black line reports the corresponding response using all speeches. Shaded areas and dotted lines denote confidence bands around the respective responses. The bottom-right tile is reserved for the legend.

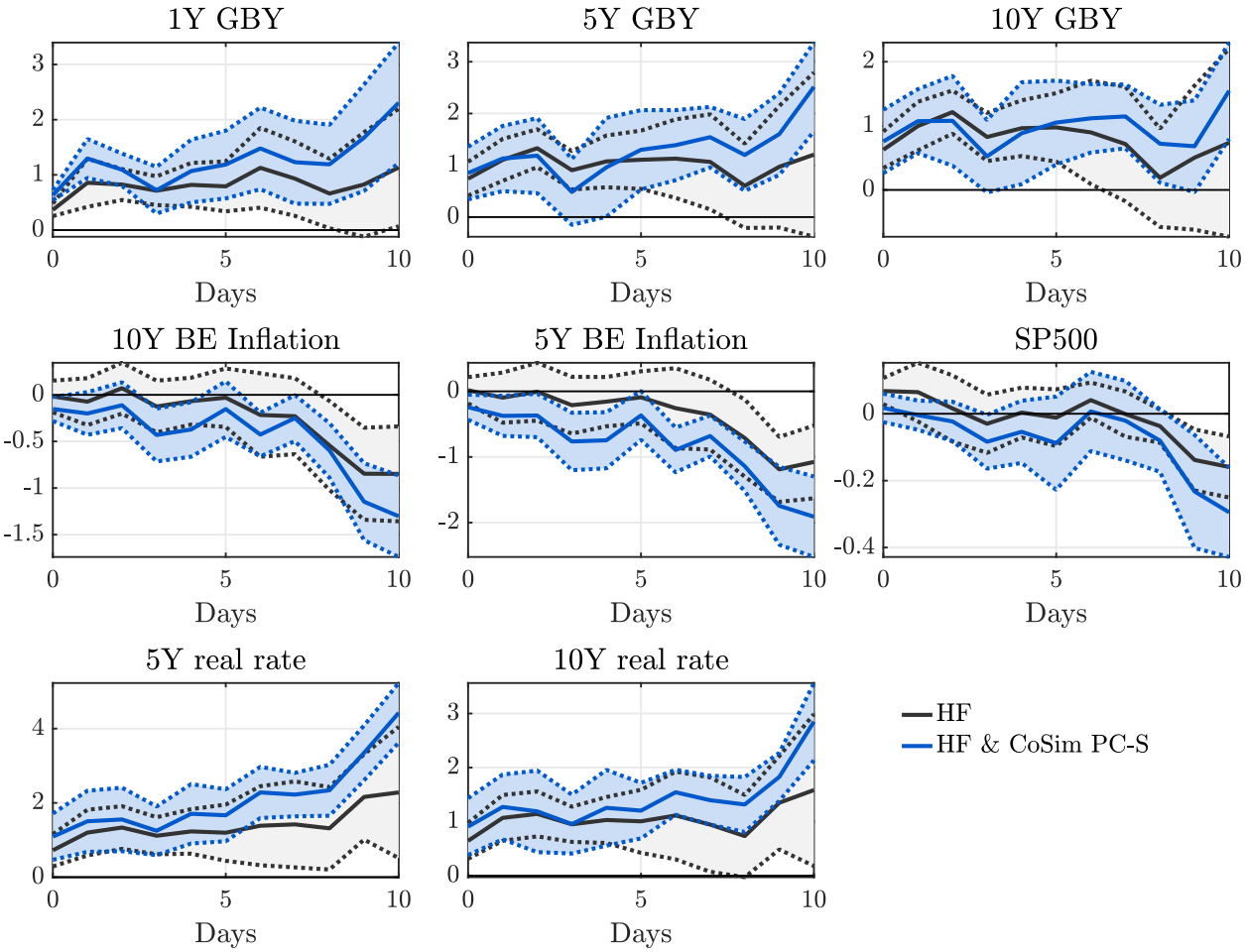
A.8 Robustness Results: Macro-News-Orthogonal Cosine Similarity

Figure A17: Financial-Market Responses to Monetary Policy Speech Surprises and Macro-News-Orthogonal Cosine Similarity ($h = 1$).



Notes: Each panel plots the estimated effect of a tightening monetary policy speech (MPS) surprise on the indicated financial variable as a function of the residuals from a regression of the baseline cosine-similarity score between the speech and the Chair's most recent post-meeting press conference on the macroeconomic-news indexes described in Subsection 2.3 (macro-news-orthogonal cosine similarity, horizontal axis). The solid blue line shows the fitted marginal effect $\beta_h + \delta_h \ell_t^{\perp}$ evaluated at $h = 1$ day after the speech; shaded areas (and dotted boundaries) report the corresponding confidence band. Black circles denote the unconditional effect estimated without interacting the MPS surprise with similarity. The legend distinguishes the unconditional (HF) effect from the similarity-conditioned effect.

Figure A18: Dynamic Responses to Speech Surprises: Macro-News-Orthogonal Cosine Similarity.



Notes: This figure reports daily local-projection responses of financial variables to a tightening monetary policy speech (MPS) surprise over horizons $h = 0, \ldots, 10$. The blue line isolates responses using only speeches with high macro-news-orthogonal cosine similarity to the Chair's most recent post-meeting press conference, as measured by $\ell_t^{\perp}$ (residual exceeding one standard deviation above its sample mean); the gray/black line reports the corresponding response using all speeches. Shaded areas and dotted lines denote confidence bands around the respective responses. The bottom-right tile is reserved for the legend.

B Equilibrium and Numerical Solution

This appendix supplies the equilibrium details and numerical implementation for Section 5.

B.1 Optimality and the Deterministic Steady-State

Let $u(C,N) = (C^{1-\sigma}-1)/(1-\sigma) - \chi N^{1+\varphi}/(1+\varphi)$. In the baseline the household maximizes $\mathbb{E}_0 \sum_t \beta^t u(C_t, N_t)$; in the extension it maximizes $\mathbb{E}_0 \sum_t \beta^t \Xi_t u(C_t, N_t)$, with the same nominal budget constraint. Writing $\Xi_t \equiv 1$ in the baseline unifies the derivation. The marginal value of nominal wealth is $\Xi_t C_t^{-\sigma}/P_t$. The bond first-order condition is therefore

$$\Xi_t C_t^{-\sigma} = \beta R_t \mathbb{E}_t \left[\frac{\Xi_{t+1} C_{t+1}^{-\sigma}}{\Pi_{t+1}} \right]. \tag{18}$$

Define $d(X) = 1$ in the baseline and $d(X) = e^{-r^*}$ in the extension. Since $\Xi_{t+1}/\Xi_t = d(X_t)$ is known at t, the real stochastic discount factor is $\beta d(X_t)(C_{t+1}/C_t)^{-\sigma}$. The preference weight cancels from the labor first-order condition because it multiplies the entire period utility flow.

For completeness, define $p_t(j) = P_t(j)/P_t$. An intermediate firm chooses its price to maximize

$$\mathbb{E}_t \sum_{k=t}^{\infty} M_{t,k} \left\{ p_k(j)^{1-\epsilon_p} Y_k - w_k p_k(j)^{-\epsilon_p} Y_k - \frac{\phi_p}{2} \left[\frac{P_k(j)}{P_{k-1}(j)} - 1 \right]^2 Y_k \right\}, \tag{19}$$

where $M_{t,k} = \beta^{k-t}(\Xi_k/\Xi_t)(C_k/C_t)^{-\sigma}$. Differentiating with respect to $P_t(j)$ and imposing symmetry gives equation (10). In particular, $e^{-r_t^*}$ discounts the continuation term in both the Euler and Rotemberg equations of the extension. This factor is required by the household ownership of firms.

At the zero-shock, zero-inflation deterministic steady-state allocation, $w = (\epsilon_p - 1)/\epsilon_p$, $R = \beta^{-1}$, and $C = Y = N = [w/\chi]^{1/(\sigma+\varphi)}$. The normalization $\chi = (\epsilon_p - 1)/\epsilon_p$ sets these three quantities to one.

B.2 Belief Updating and State Transitions

The signals observed at date t determine agents' posterior distribution of the innovations realized at $t+1$. The state transition therefore has two steps. First, the innovations are drawn from the current posterior and the fundamental states evolve. Second, a new similarity realization and new signals determine the posterior mean carried into the next period.

Similarity is independently drawn each period from $F_\zeta = \text{Beta}(a_\zeta, b_\zeta)$. Given its empirical mean $m = \mathbb{E}[\zeta]$ and variance $s^2 = \text{Var}(\zeta)$, the Beta parameters are

$$a_\zeta = m\left[\frac{m(1-m)}{s^2} - 1\right], \qquad b_\zeta = (1-m)\left[\frac{m(1-m)}{s^2} - 1\right]. \tag{20}$$

We use $m = 0.4327978104$ and $s = 0.1494428792$ in both models.

Baseline. After observing the two signals, agents' posterior for the future policy innovation is

$$\varepsilon^\theta_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(\mu_t, V_B(\zeta_t))\,.$$

Let u be standard normal. A conditional realization of this innovation can be written as

$$\varepsilon^\theta_{t+1} = \mu_t + \sqrt{V_B(\zeta_t)}\,u.$$

The policy disturbance therefore evolves according to

$$\theta_{t+1} = \rho_\theta \theta_t + \sigma_\theta\left[\mu_t + \sqrt{V_B(\zeta_t)}\,u\right].$$

At date $t+1$, a new similarity realization $z' \sim F_\zeta$ is drawn independently. Before observing the particular values of the next signals, their posterior mean has conditional distribution

$$\mu_{t+1} \mid \zeta_{t+1} = z' \sim \mathcal{N}(0, 1 - V_B(z'))\,.$$

Hence, letting v be an independent standard-normal variable, the complete transition from $X_t = (\theta_t, \zeta_t, \mu_t)$ can be written as

$$\begin{aligned} \theta_{t+1} &= \rho_\theta \theta_t + \sigma_\theta\left[\mu_t + \sqrt{V_B(\zeta_t)}\,u\right], \\ \zeta_{t+1} &= z', \qquad\qquad u, v \sim \mathcal{N}(0,1), \qquad z' \sim F_\zeta. \\ \mu_{t+1} &= \sqrt{1 - V_B(z')}\,v, \end{aligned} \tag{21}$$

The variables u, v, and z' are mutually independent. The first normal draw realizes the innovation described by the current signals, whereas the second generates the posterior mean associated with the next period's signals.

Demand-news extension. In the extension, agents form a joint posterior over the future policy and demand innovations:

$$\boldsymbol{\varepsilon}_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(\boldsymbol{\mu}_t, V_R(\zeta_t)), \qquad \boldsymbol{\varepsilon}_{t+1} = \begin{pmatrix} \varepsilon^{\theta}_{t+1} \\ \varepsilon^{r}_{t+1} \end{pmatrix}.$$

Let $L_V(\zeta)$ be a matrix square root of the posterior covariance:

$$L_V(\zeta) L_V(\zeta)' = V_R(\zeta).$$

A conditional realization of the two innovations is then

$$\boldsymbol{\varepsilon}_{t+1} = \boldsymbol{\mu}_t + L_V(\zeta_t)\boldsymbol{u}, \qquad \boldsymbol{u} \sim \mathcal{N}(0, I_2).$$

The two fundamental states evolve as

$$\theta_{t+1} = \rho_\theta \theta_t + \sigma_\theta \varepsilon^{\theta}_{t+1}, \qquad r^*_{t+1} = \rho_r r^*_t + \sigma_r \varepsilon^{r}_{t+1}.$$

Conditional on next period's similarity z', the next posterior mean has covariance $I_2 - V_R(z')$. Let $L_M(z')$ satisfy

$$L_M(z') L_M(z')' = I_2 - V_R(z').$$

The complete transition from $X_t = (\theta_t, r^*_t, \zeta_t, \boldsymbol{\mu}_t)$ is therefore

$$\begin{aligned}
\boldsymbol{\varepsilon}_{t+1} &= \boldsymbol{\mu}_t + L_V(\zeta_t)\boldsymbol{u}, \\
\theta_{t+1} &= \rho_\theta \theta_t + \sigma_\theta \varepsilon^{\theta}_{t+1}, \\
r^*_{t+1} &= \rho_r r^*_t + \sigma_r \varepsilon^{r}_{t+1}, \qquad \boldsymbol{u}, \boldsymbol{v} \sim \mathcal{N}(0, I_2), \qquad z' \sim F_\zeta. \\
\zeta_{t+1} &= z', \\
\boldsymbol{\mu}_{t+1} &= L_M(z')\boldsymbol{v},
\end{aligned} \tag{22}$$

The vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ and the similarity draw z' are mutually independent. At singular endpoints, we use positive-semidefinite matrix square roots instead of Cholesky factors.

Transition kernels and conditional expectations. Equations (21) and (22) define the transition kernels $Q_B(\cdot \mid X)$ and $Q_R(\cdot \mid X)$ for the baseline and the extension. Formally, for model $\ell \in \{B, R\}$,

$$Q_\ell(A \mid X) = \Pr(X_{t+1} \in A \mid X_t = X), \tag{23}$$

where X_{t+1} is generated by the corresponding transition above. For any function g, its conditional expectation is

$$\mathbb{E}_\ell[g(X_{t+1}) \mid X_t = X] = \int g(X')\, Q_\ell(dX' \mid X). \tag{24}$$

Numerically, this integral averages over the current posterior innovation, next period's similarity, and the posterior mean generated by next period's signals. Importantly, posterior variances are evaluated at the relevant current or future realization of similarity, rather than at mean similarity.

B.3 Projection with Time Iteration

We approximate $\log \mathcal{C}(X)$ and $\log \mathcal{P}(X)$ with total-degree Chebyshev polynomials on a bounded state domain.[5] Given these two policies, define $D(X) = 1 - \phi_p[\mathcal{P}(X) - 1]^2/2$ and recover

$$\mathcal{Y}(X) = \mathcal{N}(X) = \frac{\mathcal{C}(X)}{D(X)}, \qquad \mathcal{W}(X) = \chi \mathcal{N}(X)^\varphi \mathcal{C}(X)^\sigma, \tag{25}$$

with $\mathcal{R}(X)$ from the Taylor rule. Admissibility requires $D(X) > 0$. The two continuation objects are

$$\begin{aligned} \mathcal{E}_1(X) &= \mathbb{E}\left[\frac{\mathcal{C}(X')^{-\sigma}}{\mathcal{P}(X')} \,\middle|\, X\right], \\ \mathcal{E}_2(X) &= \mathbb{E}\left[\mathcal{C}(X')^{-\sigma}[\mathcal{P}(X') - 1]\mathcal{P}(X')\mathcal{Y}(X') \,\middle|\, X\right]. \end{aligned} \tag{26}$$

For fixed continuation values, current consumption and inflation solve

$$\begin{aligned} 0 &= -\sigma \log C - \log \beta - \log R - \log d(X) - \log \mathcal{E}_1(X), \\ 0 &= \phi_p(\Pi - 1)\Pi - [1 - \epsilon_p + \epsilon_p w] - \beta d(X)\phi_p \frac{C^\sigma}{Y}\mathcal{E}_2(X). \end{aligned} \tag{27}$$

Thus, the numerical equilibrium consists of two nonlinear functional equations plus the four static restrictions. It implements the same equilibrium conditions in both models, with the additional preference factor $d(X) = e^{-r^*}$ in the extension.

Starting from the deterministic reference policies, each time-iteration step evaluates the continuation objects at the collocation nodes, solves the two current-period equations, and

[5] A total-degree-p basis contains products $\prod_{i=1}^d T_{k_i}(x_i)$ satisfying $\sum_{i=1}^d k_i \le p$, where T_{k_i} is the Chebyshev polynomial of degree k_i in state i. This gives 20 basis terms for the three-state, degree-three baseline and 126 terms for the five-state, degree-four extension. The collocation grid itself remains a tensor-product grid.

projects the updated log policies onto the polynomial basis by least squares. Coefficients are damped by one half.

The baseline approximation domain is $\theta \in [-0.02, 0.02]$, $\zeta \in [0, 1]$, and $\mu \in [-5, 5]$. The extension uses the same domains for θ and ζ, together with $r^* \in [-0.035, 0.035]$ and $\mu^\theta, \mu^r \in [-3, 3]$. Tensor products of Chebyshev roots form the collocation grids. Gaussian states retain their unbounded theoretical support; these intervals delimit only the approximation domain, and the policy functions are extrapolated when a state falls outside it.

Conditional expectations are evaluated using Gaussian quadrature and Gauss–Legendre nodes for similarity, reweighted by the calibrated Beta density. We assess accuracy at both the collocation nodes and independently drawn off-grid states. In the baseline, the maximum off-grid Euler and pricing residuals are 1.01×10^{-8} and 5.50×10^{-7}, respectively. In the extension, they are 1.38×10^{-6} and 6.01×10^{-5}. Less than 0.1 percent of observations in the extension's stationary simulation fall outside the approximation domain, and none do so in the baseline simulation.

B.4 Computing the Profiles Reported in the Text

For each value of similarity, we evaluate the solved equilibrium at a communication-news state and at a zero-message benchmark. In the baseline, these states are

$$X_B^N(\zeta) = \left(0, \zeta, \frac{2\zeta}{1+\zeta}\right), \qquad X_B^0(\zeta) = (0, \zeta, 0). \tag{28}$$

The first component is the current policy disturbance, the second is similarity, and the third is the posterior mean of the future policy innovation. Hence, $X_B^N(\zeta)$ corresponds to $s_t^C = s_t = 1$, while $X_B^0(\zeta)$ corresponds to $s_t^C = s_t = 0$.

In the demand-news extension, let

$$\boldsymbol{\mu}_R^N(\zeta) = K(\zeta) \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \mu_R^{\theta,N}(\zeta) \\ \mu_R^{r,N}(\zeta) \end{pmatrix}. \tag{29}$$

The corresponding states are

$$X_R^N(\zeta) = \left(0, 0, \zeta, \mu_R^{\theta,N}(\zeta), \mu_R^{r,N}(\zeta)\right), \qquad X_R^0(\zeta) = (0, 0, \zeta, 0, 0). \tag{30}$$

The current policy and demand states are set to their deterministic steady-state values. In particular, $r_t^* = 0$ means that the preference-driven natural-rate component is at its steady-state value, not that the level of the natural rate is zero. Within each comparison, similarity

is held fixed, so the news state and its benchmark have the same posterior covariance and differ only in their posterior means.

For model $\ell \in \{B, R\}$, define the outcome map

$$\mathbf{\Psi}_\ell(X) = \begin{pmatrix} \int \log \mathcal{R}(X')\, Q_\ell(dX' \mid X) \\ \log \mathcal{P}(X) \\ \int \log \mathcal{P}(X')\, Q_\ell(dX' \mid X) \end{pmatrix}. \tag{31}$$

Its components are the expected next-quarter short rate $\mathbb{E}_t i_{t+1}$, current inflation π_t, and expected next-quarter inflation $\mathbb{E}_t \pi_{t+1}$. The response of outcome j is

$$r_{\ell,j}(\zeta) = 10^4 \left[\Psi_{\ell,j}\big(X_\ell^N(\zeta)\big) - \Psi_{\ell,j}\big(X_\ell^0(\zeta)\big)\right], \qquad \widetilde{r}_{\ell,j}(\zeta) = \frac{r_{\ell,j}(\zeta)}{r_{\ell,Ei}(\overline{\zeta})}. \tag{32}$$

The denominator is fixed as similarity varies and is computed separately for the two models. The nonlinear response is evaluated at 201 equally spaced values of ζ. In each panel, the blue line reports this nonlinear profile and the black circle reports the similarity-averaged response $\overline{r}_{\ell,j}$ at mean similarity.

C Communication Experiments and Dynamic Responses

The main-text figures report impact responses to communication news. To illustrate their timing, we also evaluate four-quarter paths. Quarter 1 starts with zero current fundamental states and signals $s_1^C = s_1 = 1$. Initial similarity takes one of three values, $\zeta_1 \in \{0.3, \overline{\zeta}, 1\}$, and equals $\overline{\zeta}$ from quarter 2 onward. The signaled policy innovation is realized in quarter 2: $\varepsilon_2^\theta = 1$. Later policy innovations and signals are zero, so $\theta_t = \sigma_\theta \rho_\theta^{t-2}$ for $t \geq 2$. In the extension, the realized natural-rate innovation is zero throughout. All responses subtract a zero-message, zero-fundamental path at the same sequence of similarity states.

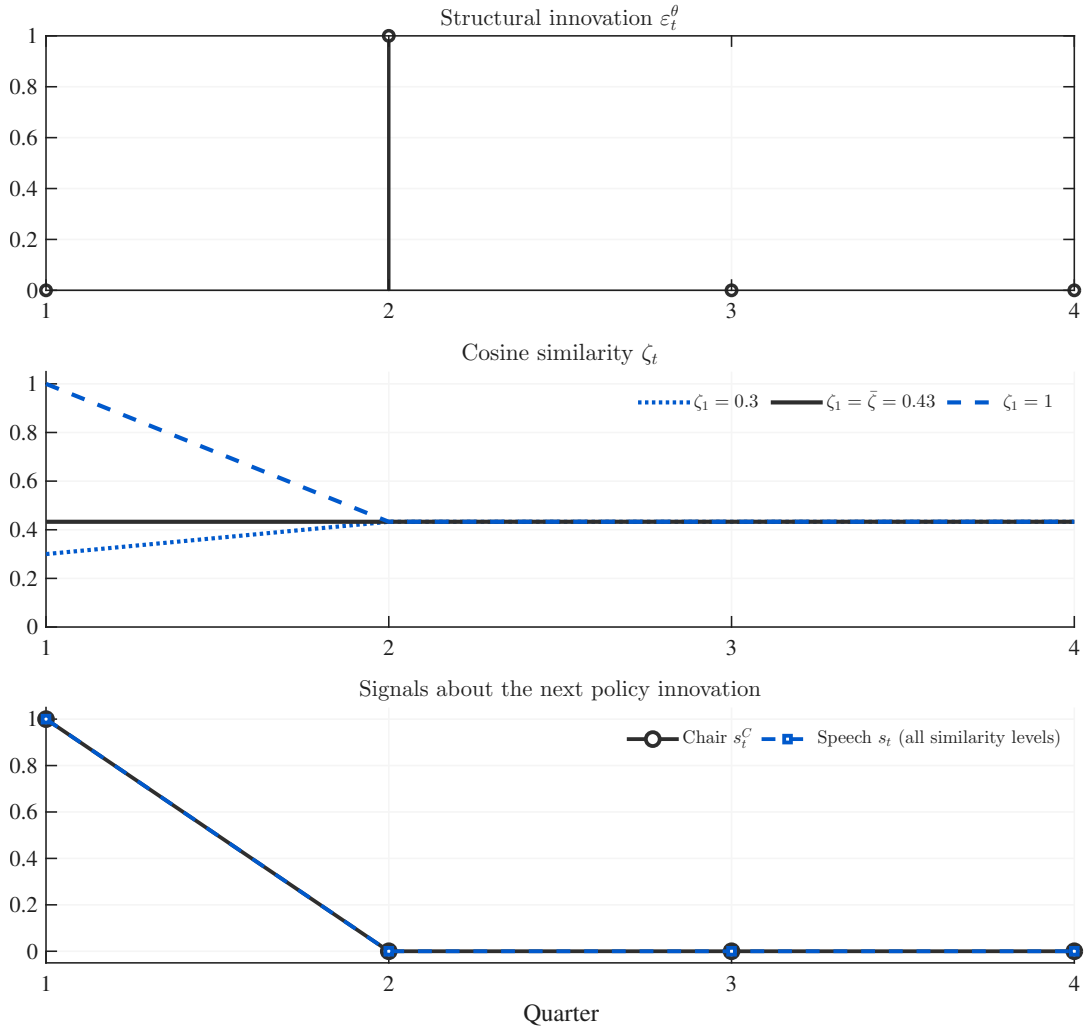
Figures A19 and A20 show the baseline. The unit messages are consistent with zero measurement errors and the future unit innovation. Higher initial similarity raises the anticipated tightening and produces a larger immediate fall in inflation. Current nominal rates initially fall through the endogenous Taylor-rule response; the positive policy innovation arrives in quarter 2. Once the innovation is realized and similarity is common, the paths coincide. The black reference is the two-signal equilibrium at usual similarity, not a no-speech counterfactual.

Figures A21 and A22 use the same observed messages in the extension. Holding $s_1 = 1$ while realizing $(\varepsilon_2^\theta, \varepsilon_2^r) = (1, 0)$ requires the speech error to satisfy $\nu_R(\zeta_1)\eta_1 = 1 - \zeta_1$ for

$\zeta_1 < 1$. The Chair error is zero. Thus, these are fixed-message comparisons with different interpretations of the news, not zero-noise paths at every similarity. At low similarity, agents initially infer favorable demand news even though the subsequent realized innovation is monetary. At $\zeta_1 = 1$, both messages reveal the policy innovation perfectly.

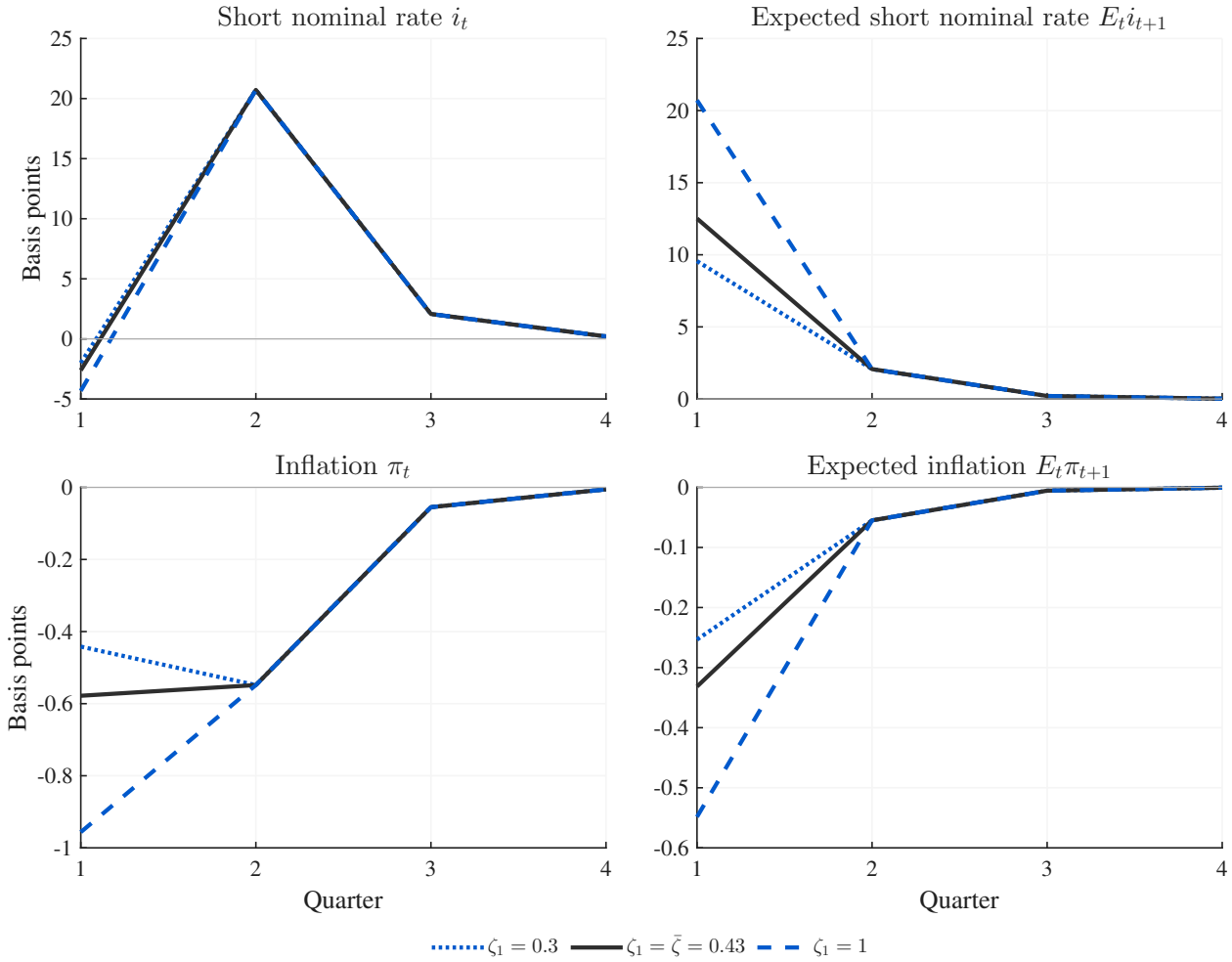
These figures are conditional scenario paths evaluated with the stochastic equilibrium policies. At every point, the expected rate and expected inflation panels still integrate over the model's conditional distribution of the next state. All panels report changes in log gross variables multiplied by 10^4, without annualization.

Figure A19: Timing of the Baseline Policy-News Experiment



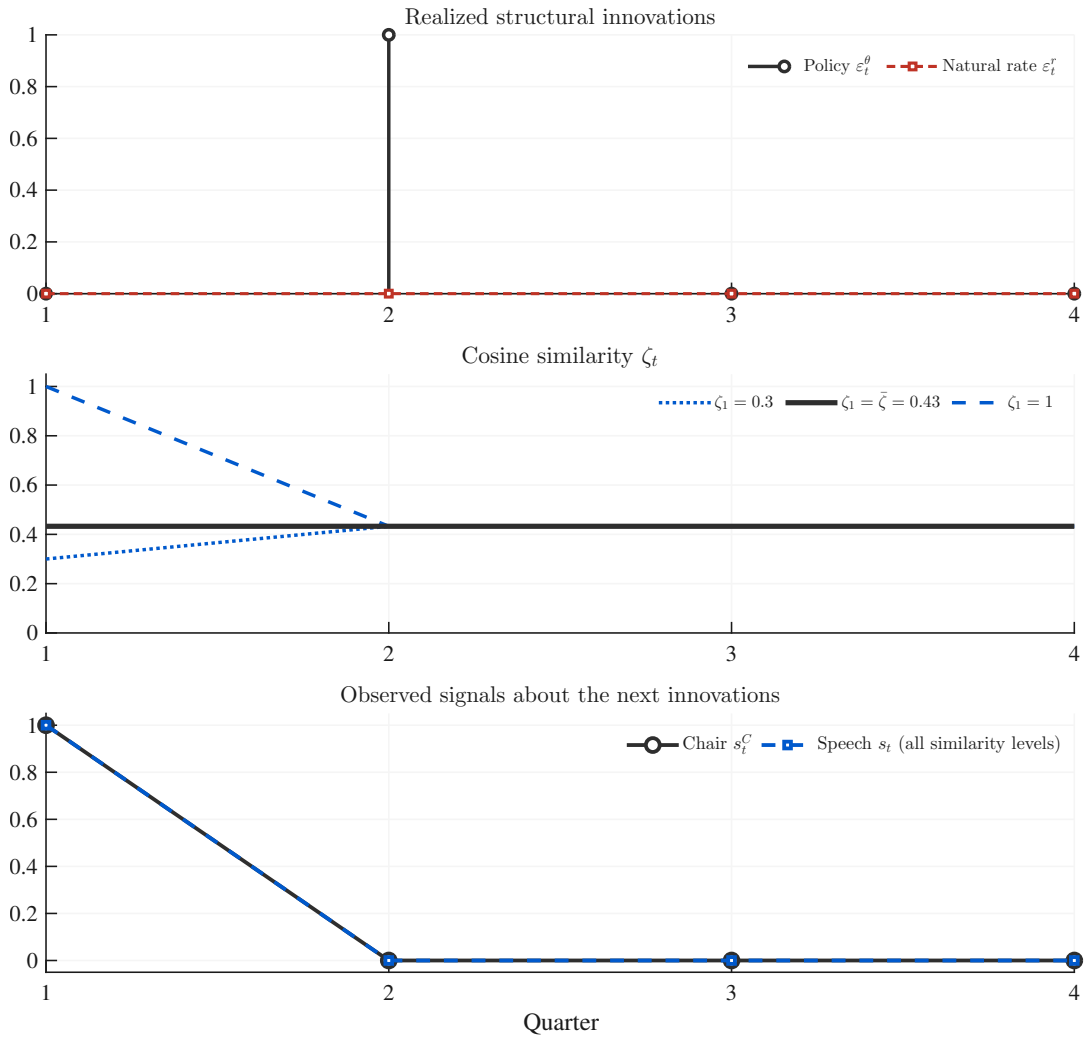
Notes: The top panel shows the standardized policy innovation, whose unit realization in quarter 2 shifts the policy rule by 25 basis points. The middle panel varies initial similarity and returns all paths to $\bar{\zeta} = 0.4328$ in quarter 2. The bottom panel shows the two unit messages received in quarter 1; subsequent messages are zero.

Figure A20: Baseline Responses to News of a Future Tightening



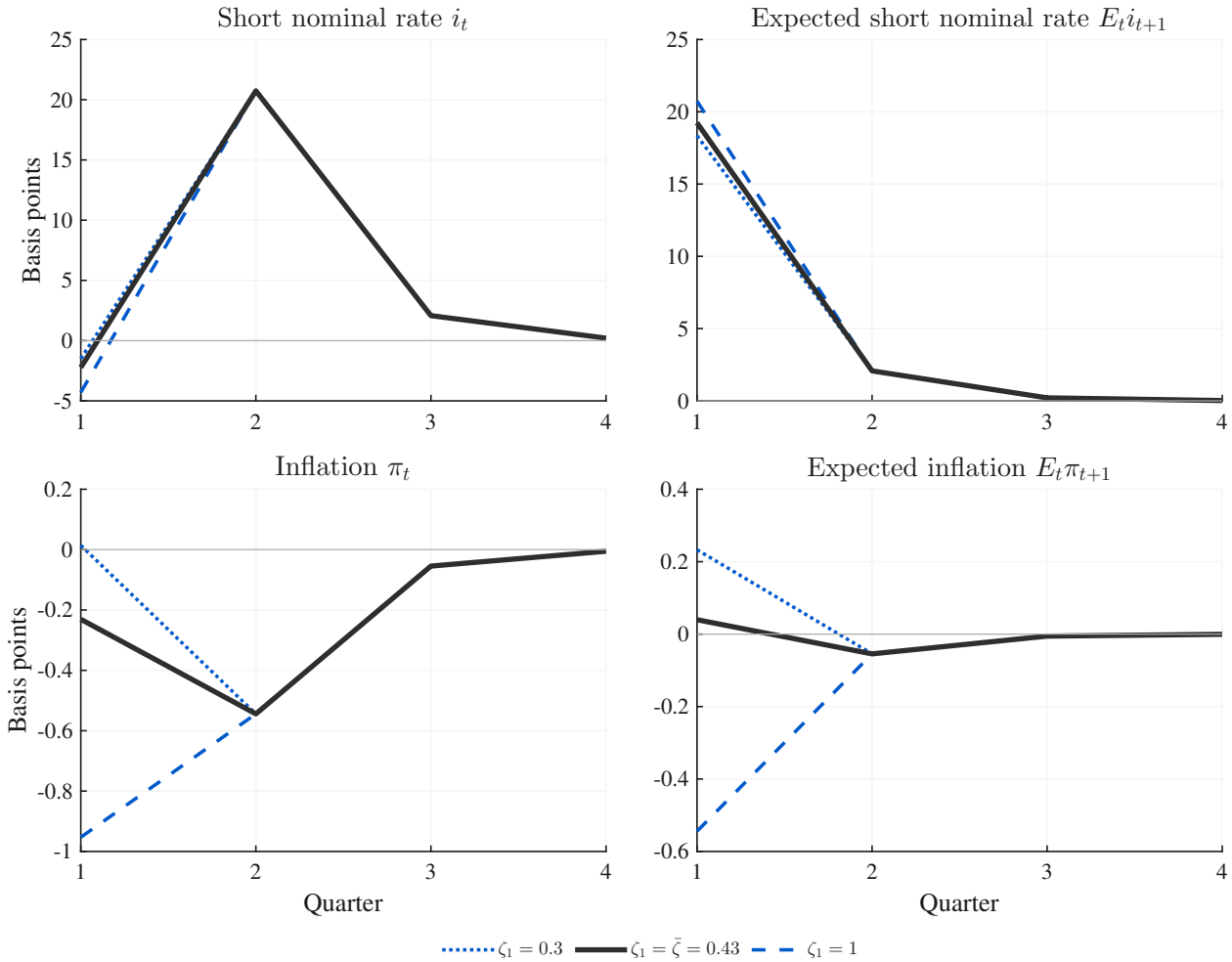
Notes: Responses to the experiment in Figure A19, relative to a zero-news path with the same similarity sequence. Units are basis points. The three lines correspond to $\zeta_1 = 0.3$, $\zeta_1 = \bar{\zeta}$, and $\zeta_1 = 1$. Expected outcomes are conditional expectations from the nonlinear equilibrium. The realized quarter-2 policy innovation is common to all paths; communication differs only in quarter 1.

Figure A21: Timing of the Experiment with Policy and Demand News



Notes: The observed messages and similarity paths are those of the baseline experiment. A unit policy innovation is realized in quarter 2 and the realized demand innovation is zero. The fixed speech message therefore entails a measurement error at $\zeta_1 < 1$. Agents infer both future innovations from the messages before those innovations are realized.

Figure A22: Dynamic Responses with Policy and Demand News



Notes: Responses to the experiment in Figure A21, in basis points relative to a zero-news path at the same similarity. Low similarity initially raises expected inflation through the demand-news component; high similarity makes the anticipated policy tightening dominant. The black line uses $\bar{\zeta} = 0.4328$. Paths coincide from quarter 2, when the common policy innovation is realized, the demand state is zero, and all paths return to usual similarity with zero subsequent messages.